

Reinforcement learning for logistics and supply chain management: Methodologies, state of the art, and future opportunities

Yimo Yan^a, Andy H.F. Chow^b, Chin Pang Ho^c, Yong-Hong Kuo^{a,d,*}, Qihao Wu^a, Chengshuo Ying^e

^a Department of Industrial and Manufacturing Systems Engineering, the University of Hong Kong, Hong Kong Special Administrative Region

^b Department of Advanced Design and Systems Engineering, City University of Hong Kong, Hong Kong Special Administrative Region

^c School of Data Science, City University of Hong Kong, Hong Kong Special Administrative Region

^d HKU Musketeers Foundation Institute of Data Science, the University of Hong Kong, Hong Kong Special Administrative Region

^e Yangtze Delta Region Academy of Beijing Institute of Technology, Jiaxing, China

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ABSTRACT

With advances in technologies, data science techniques, and computing equipment, there has been rapidly increasing interest in the applications of reinforcement learning (RL) to address the challenges resulting from the evolving business and organisational operations in logistics and supply chain management (SCM). This paper aims to provide a comprehensive review of the development and applications of RL techniques in the field of logistics and SCM. We first provide an introduction to RL methodologies, followed by a classification of previous research studies by application. The state-of-the-art research is reviewed and the current challenges are discussed. It is found that Q-learning (QL) is the most popular RL approach adopted by these studies and the research on RL for urban logistics is growing in recent years due to the prevalence of E-commerce and last mile delivery. Finally, some potential directions are presented for future research.

1. Introduction

Logistics and supply chain management (SCM) can be regarded as the processes of moving products and goods from production (suppliers) to consumption (customers). The problems in logistics and SCM have always been an important subject in the civil and military domains since the 20th century BC (Ghiani et al., 2013). Since the last century, with the introduction to operational research and computers, various optimisation models and algorithms have been proposed and applied, such that the problems in logistics and SCM can be solved systematically and scientifically. More recently, with the advancements in data and computer sciences, we have been witnessing the growth in applications of reinforcement learning (RL) for solving decision problems. Typically, these decision problems involve large state and decision spaces with uncertainties in the system. Due to the complexities of logistics and supply chain (SC) systems, RL has also gained greater popularity in this domain in recent years.

RL is a computing approach to solve sequential decision-making problems. At each decision stage, the RL controller derives an admissible action via a policy function based on the prevailing state estimated by a value function to maximise the corresponding rewards gained. In the context of transport and logistics, the states of a system are usually characterised by the statuses of vehicles, customer demands and their locations, time windows, and inventory levels. Meanwhile, the actions can be vehicles' routes, order quantities, or prices of delivery services. The rewards can be related to the profits generated, distance moved, energy consumed, or

* Corresponding author.

E-mail address: yhkuo@hku.hk (Y.-H. Kuo).

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Table 1

Table of abbreviations.

A2C	Advantage Actor-Critic	A3C	Asynchronous Advantage Actor-Critic
AC	Actor-Critic	ADP	Adaptive Dynamic Programming
ANN	Artificial Neural Network	CNN	Convolutional Neural Network
CVRP	Capacitated Vehicle Routing Problem	DDPG	Deep Deterministic Policy Gradient
DNN	Deep Neural Network	DP	Dynamic Programming
DQN	Deep Q Network	DRL	Deep Reinforcement Learning
EV	Electric Vehicle	GA	Genetic Algorithm
GAN	Graph Attention Network	GCN	Graph Convolutional Network
GNN	Graph Neural Network	HRL	Hierarchical Reinforcement Learning
IRL	Inverse Reinforcement Learning	LA	Locating Agent
LSTM	Long Short Term Memory	M(S)VRPPD	Multi (Single) Vehicle Routing Problem with Pickup and delivery
MARL	Multi-Agent Reinforcement Learning	MDP	Markov Decision Process
MILP	Mixed-Integer Linear Programming	MINLP	Mixed-Integer Non-Linear Program
NN	Neural Network	PDP	Pickup and Delivery Problem
PG	Policy Gradient	PPO	Proximal Policy Optimisation
QL	Q-learning	ReLU	Rectified Linear Unit
RL	Reinforcement Learning	RLSO	Reinforcement Learning and Stochastic Optimisation
RNN	Recurrent Neural Network	SARSA	State–Action–Reward–State–Action
SC	Supply Chain	SCM	Supply Chain Management
SMART	Semi-Markov Average Reward Technique	SRDQN	Shaped Reward Deep Q Network
TA	Transporting Agent	TD	Temporal-Difference
TSP	Travelling Salesman Problem	UAV	Unmanned Aerial Vehicle
UCC	Urban Consolidation Centre	UV	Unmanned Vehicle
VMI	Vendor Managed Inventory	VRP	Vehicle Routing Problem
VRPPD	Vehicle Routing Problem with Pick-up and Delivering	VRPTW	Vehicle Routing Problem with Time Windows

other performance indicators. The development and application of RL could play an important role in the future development of smart logistics and SC systems.

This paper aims to provide a comprehensive review of the development and applications of RL techniques in the field of logistics and SCM. We define the applications for logistics and SCM as the movements of products and goods and the relevant processes to satisfy consumer demands, where the research on traffic and transport of people is excluded in this literature review. For literature reviews of traffic or transport of people, we refer the reader to [Farazi et al. \(2020\)](#) and [Qin et al. \(2021\)](#). To the best of our knowledge, this is among the first effort to provide a comprehensive review of RL applications in the field of logistics and SCM. The goal of this review is to introduce to the logistics research community the potential and state of the art of RL, and hence identify promising research directions. The specific research questions we aim to answer through this study are:

- Which applications in logistics and SCM are RL methods more promising to be adopted?
- Which RL methods are more appropriate in solving a certain problem?
- What are the challenges and future opportunities of RL applications for logistics and SCM?

This paper requires the use of abbreviations, which are presented in [Table 1](#).

Our paper is organised as follows. In [Section 2](#), we introduce the core elements of RL. In [Section 3](#), we classify and review the existing studies into different categories by application. [Section 4](#) presents the current research trends and future directions. Finally, we conclude this paper in [Section 5](#).

2. Methodologies

In this section, we will provide the basics of the most popular methodologies used in RL approaches for logistics and supply chain applications.

2.1. Markov decision process

An RL framework is typically built upon a Markov Decision Process (MDP), which can be formulated as:

$$\max_{\pi} \lim_{T \rightarrow \infty} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t | s_0 = s, \pi \right], \quad (1)$$

in which π is a policy function that derives corresponding admissible actions $\{a_0, a_1, \dots\}$ to be implemented at each decision stage t over an infinite horizon T based on the associated states $\{s_0, s_1, \dots\}$; $\{s_0, s_1, \dots\}$ are the state variables of the system over each stage t with s_0 being the initial state. Given the initial state $s_0 = s$, MDP (1) aims to determine an optimal policy function π^* with which the total expected rewards to be collected over T can be maximised. The essential property of an MDP is that, given the current state s_t , the next state s_{t+1} of the system is independent of the previous states $\{s_0, s_1, \dots, s_{t-1}\}$ ([Haydari and Yilmaz, 2022](#)).

To ensure convergence of the MDP (1), a discount factor γ' , where $\gamma \in (0, 1)$, is associated with each future reward r_t (Bertsekas, 2019, 2021). This discount factor determines how the reward at future stage t is valued (Powell, 2011).

An optimal action a_t^* under a policy π can be computed iteratively over each stage t from the optimality condition:

$$a_t^* = \arg \max_{a_t} \mathbb{E}_{\pi} [r_t + \gamma V_{\pi}(s_{t+1})], \quad (2)$$

in which r_t is the immediate reward induced by action a_t at the current stage t ; $V_{\pi}(\cdot)$ is known as the value function, which represents the total optimal reward from stage t onward given the current state s_t and action a_t . The key for solving a_t^* in Eq. (2) lies in the calculation of the value function $V_{\pi}(s)$. This value function can be written recursively following the well-established Bellman equation (Powell, 2011; Bertsekas, 2019) as:

$$V_{\pi}(s_t) = \max_{a_t} \mathbb{E}_{\pi} [r_t + \gamma V_{\pi}(s_{t+1})] \quad (3)$$

Solving Eq. (2) for a_t with iteratively updating V via Eq. (3) encounters several major computational challenges. First, the recursive Eq. (3) reveals that the value function V_{π} will need to be calculated backwards in stage due to the existence of uncertainties (Powell, 2011; Bertsekas, 2019). Moreover, obtaining the value function in Eq. (3) would need to visit all reachable states with the evaluation of all admissible actions. Consequently, calculating the true value of V_{π} can be computationally expensive even for medium-sized problems (Powell, 2011). The computational issue identified herein is known as the “curse of dimensionality” in the literature of dynamic programming (DP) (Bertsekas, 2019), which is the most reported reason that many dynamic optimisation problems are computationally intractable when being applied to real-world problems.

2.2. Reinforcement learning

As aforementioned, deriving an exact solution for the MDP is often infeasible and computationally intractable due to the “curses of dimensionality” (Bertsekas, 2019). This leads to various approaches for solving the MDP formulations with the use of approximation techniques under an RL framework. There are two general kinds of approximations adopted for solving the MDP: value-based approximation and policy-based approximation. In terms of implementation, we can further classify them into model-based implementation and model-free implementation, as well as offline implementation and online implementation.

2.2.1. Value-based approximation

A value-based approximation estimates the value function $V_{\pi}(s_t)$ at current state s_t with some other function $\tilde{V}_{\pi}(s_t)$. Optimality Condition (2) hence will become

$$\tilde{a}_t^* = \arg \max_{a_t} \mathbb{E}_{\pi} [r_t + \gamma \tilde{V}_{\pi}(s_{t+1})], \quad (4)$$

in which $\tilde{a}_t^*$ is an suboptimal action at current t subject to the choices and settings of $\tilde{V}_{\pi}$. It is common to express the expectation on the right-hand-side of Eq. (4) as (Bertsekas, 2019):

$$Q(s_t, a_t) = \mathbb{E}_{\pi} [r_t + \gamma \tilde{V}_{\pi}(s_{t+1})], \quad (5)$$

where $Q(s_t, a_t)$ is known as the Q -value given the associated state and action are s_t and a_t respectively, with which Optimality Condition (4) can be further rewritten as:

$$\tilde{a}_t^* = \arg \max_{a_t} Q(s_t, a_t), \quad (6)$$

Various value-based approximations have been proposed to solve the original MDP via Eq. (6) with iteratively approximating and updating the Q -value in Eq. (5). Details of the value-based approximations and Q -value approximations will be introduced in latter subsections.

2.2.2. Policy-based approximation

Policy-based approximation is an alternative to value-based approximation for solving MDP, where we make approximation $\tilde{\pi}$ to the policy function π instead of the value function V as

$$\tilde{a}_t^* = \tilde{\pi}(s_t, \phi)$$

which derives a suboptimal action a_t based on the prevailing state s_t and parameters set ϕ . An advantage of this policy-based approximation approach over the value-based approximation approach is that, once the parameters set ϕ is determined, the corresponding action a_t can be derived based on the prevailing state s_t without solving Maximisation Problem (2).

2.2.3. Model-based and model-free implementations

Regardless of whether a value-based or a policy-based approximation is adopted, an RL framework can be implemented via either a model-based approach or a model-free approach. Following Bertsekas (2019), we consider a model-based implementation having an assumption that the underlying state transition and its probability distribution can be expressed by explicit functions or models. For more details, we refer the reader to a comprehensive survey on model-based implementations by Moerland et al. (2020). In contrast, the state transition and expected reward values are calculated by Monte Carlo simulations without having an explicit mathematical model in a model-free implementation. Generally, model-based algorithms are more computationally expensive, but perform better with exploration.

2.2.4. Offline and online methods

An important aspect to consider when designing an RL framework is whether the approximations are calculated offline (i.e., before the decision process starts) or online (i.e., during the decision process). The advantage of an offline calculation method is the computational efficiency because no extra training or update for the approximations is needed once the decision or control process starts. Examples of offline implementations include various regression and Artificial Neural Network (ANN)-based value function or policy function approximation. With online methods, the approximations are calculated and updated every time when a state is revealed and a decision is made. Despite the computational effort required for online training, the advantage of an online method over an offline one is its adaptiveness to prevailing system conditions, in particular when there is a significant change in system characteristics before and after the control process starts. Examples of online implementations include conventional rollout or model prediction control settings (Bertsekas, 2019; Sutton and Barto, 2018).

2.2.5. Temporal-difference learning

Temporal-difference (TD) learning is a core component in RL (Sutton and Barto, 2018). This is a method for updating the approximations (value-based or policy-based) performed in an RL framework via successive collections of state transitions and rewards. TD learning can be carried out in two ways: on-policy and off-policy methods. Following Sutton and Barto (2018), on-policy methods attempt to evaluate and improve the policy function π , that is currently used to derive current decisions. In contrast, off-policy methods aim to evaluate and update a policy function π , that is different from the one currently used to derive the decisions. The difference between the two training methods can be best illustrated by State–Action–Reward–State–Action (SARSA) (on-policy) and Q -learning (QL) (off-policy) algorithms, respectively. Both SARSA and QL algorithms update the approximations by comparing the value of the policy function iteratively at the current stage, and the value after the transition to the next stage associated with an immediate reward collected.

For SARSA as an on-policy TD learning algorithm, given the current Q -value with state–action pair (s_t, a_t) , it is updated with the SARSA rule, following each state transition from t to $t + 1$ as:

$$Q(s_t, a_t)_{new} := Q(s_t, a_t) + \alpha [r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)] \quad (7)$$

in which α is the learning rate defined by the user.

For QL as an off-policy TD learning algorithm, the RL agent aims to determine an optimal policy function that is different from the policy function currently in use.

$$Q(s_t, a_t)_{new} := Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]. \quad (8)$$

A subtle but important difference between the SARSA and QL algorithms lies in the terms $\gamma Q(s_{t+1}, a_{t+1})$ in Eq. (7) and $\gamma \max_a Q(s_{t+1}, a)$ in Eq. (8). In case of QL off-policy learning, the Q -value directly approximates the optimal action $\max_a Q(s_{t+1}, a)$, while SARSA approximates the current action a_{t+1} . It has been shown that the use of an off-policy algorithm simplifies the analysis of a learning algorithm and guarantees convergence for the learning algorithm (Sutton and Barto, 2018). Compared with an off-policy algorithm, however, the advantage of using on-policy training is that it could achieve better training performance via a more stable training process by optimising the policy that is used to explore the state and decision spaces directly (Sutton and Barto, 2018).

2.2.6. Exploration versus exploitation

With the approximation and TD learning algorithm incorporated (see Eqs. (7) and (8)), a question to address is whether we should simply implement the optimal decision derived from the current approximation settings (i.e., exploitation), or we should intentionally implement an action different to further refine and improve the current approximation (i.e., exploration) during the TD learning process. This question is known in the RL literature as the “exploration vs. exploitation” problem (Powell, 2011).

There are various ways of implementing and balancing exploration and exploitation in the decision-making process under an RL framework. One of the most widely used approaches is the ϵ -greedy search approach, which chooses between exploration and exploitation actions randomly by a predefined probability ϵ . Given a small probability ϵ , the ϵ -greedy search can be implemented as (Sutton and Barto, 2018):

$$a_t = \begin{cases} \arg \max_a Q(s_t, a) & \text{if } X > \epsilon, \\ a' & \text{otherwise.} \end{cases}$$

where $X \sim U(0, 1)$, $Pr\{a' = a\} = 1/|\mathcal{A}|$, $\forall a \in \mathcal{A}$, and $\mathcal{A}$ is the action space.

2.3. Variants of reinforcement learning

In this subsection, we will present some variants of RL.

2.3.1. Multi-agent reinforcement learning

The RL methods discussed previously are managed by a single control agent. A Multi-Agent Reinforcement Learning (MARL) framework consists of multiple control agents who can learn and make decisions individually (Lowe et al., 2017; Bertsekas, 2021). Consider an MARL with N agents, the MDP for each individual agent n , where $n = 1, 2, \dots, N$, in this N -agent MARL can be written as

$$\max_{\pi_n} \lim_{T \rightarrow \infty} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_{(n)t} | s_0 = s, \pi \right],$$

in which π_n is the local policy to be implemented by agent n , $r_{(n)t}$ is the reward gained by agent n over stage t , π is a vector denoting the joint policies of all N agents. The corresponding Q -function can then be formulated within the framework as

$$Q_{\pi_n}(s_t, \mathbf{a}) = r_{(n)t}(s_t, \mathbf{a}) + \gamma \mathbb{E}_{\pi} [V_{\pi(n)}(s_{t+1})],$$

where $\mathbf{a}$ denotes the joint actions of all agents and $V_{\pi(n)}$ is the value function for agent n with joint policy π .

In a MARL system, the interaction between agents can be cooperative, competitive, or a combination of both (Lowe et al., 2017). The collection of actions of all N agents in a MARL system can result in a dynamic game (Yang et al., 2020). The reader may refer to Panait and Luke (2005) and Busoniu et al. (2008) for surveys of multi-agent learning approaches and applications.

The simplest approach to developing a MARL system is to adopt independently learning agents. Nevertheless, it has been shown that achieving stability could be challenging if one allows each agent to act and learn independently. As discussed in Lowe et al. (2017), previous approaches to address this include incorporating other agent's policy parameters to the Q -function (Tesauro, 2004), explicitly adding the iteration index to the replay buffer, or using importance sampling (Foerster et al., 2017). In view of the stability issue, Lowe et al. (2017) and Foerster et al. (2018) propose a centralised critic-decentralised actor for stabilising the training process in a MARL system.

2.3.2. Inverse reinforcement learning

Inverse Reinforcement Learning (IRL) has an entirely different objective from other RL approaches. It is suited to an environment which is not well defined. Agents can learn from the actions of others. Some studies regard IRL as a type of imitation learning. However, a traditional imitation learning technique directly learns the actions. On the other hand, IRL involves two stages, deriving the reward function from experts' and/or optimal actions. This method allows rewards to be derived, so that humans can examine the reasoning of experts' actions. From our literature search, only one study adopts this technique; therefore, only the technique applied in that study, maximum entropy deep inverse RL, is presented in this section. There are also other popular approaches in imitation learning and IRL, such as Guided Cost Learning and Generative Adversarial Imitation Learning.

The derivation of the reward function depends on experts' actions and states, denoted by $\tau = (s_1, \mathbf{a}_1, \dots, s_n, \mathbf{a}_n)$, which is a trajectory of state-action pairs. The reward function $R_{\psi}(\tau)$, modelled as a Neural Network (NN) with parameter ψ learns from expert demonstrations $\{\tau_1, \dots, \tau_K\}$, where K is number of demonstrations and $k = 1, 2, \dots, K$. The probability $p(\tau_k | \psi)$ of having trajectory τ_k given the NN parameter ψ is shown as (Ziebart et al., 2008):

$$p(\tau_k | \psi) = \frac{1}{Z} \exp(R_{\psi}(\tau_k))$$

where

$$Z = \int_{\tau} p(\tau) \exp(R_{\psi}(\tau)) d\tau$$

Here, $p(\tau)$ is the probability of having trajectory τ as an observation, and Z acts as a normalisation term. This normalisation term prevents the NN from assigning huge rewards instead of maximising the likelihood of having the observations. The aim of IRL is to maximise the probability of having such actual observations $\{\tau_1, \dots, \tau_K\}$ with the optimal set of NN parameters. This problem is hence converted into a maximum log-likelihood problem:

$$\psi^* = \arg \max_{\psi} \mathcal{L}(\psi), \quad \text{where } \mathcal{L}(\psi) := \sum_{k=1}^K \log p(\tau_k | \psi)$$

where $\mathcal{L}(\psi)$ is the loss function of parameter ψ and $\{\tau_1, \dots, \tau_K\}$ are the observations from experts' demonstrations.

Taking the derivative of the maximum likelihood expression with respect to ψ , one can find the direction along which an optimiser can descend. The first derivative of the loss function is as follows.

$$\nabla_{\psi} \mathcal{L}(\psi) = \frac{1}{K} \sum_{k=1}^K \nabla_{\psi} R_{\psi}(\tau_k) - \frac{1}{Z} \int_{\tau} p(\tau) \exp(R_{\psi}(\tau)) \nabla_{\psi} R_{\psi}(\tau) d\tau$$

The reward NN model is updated iteratively until it converges. At the second stage, this model is served as the reward function. A standard RL model can be trained to derive the optimal actions (Ziebart et al., 2008).

2.3.3. Hierarchical reinforcement learning

Hierarchical Reinforcement Learning (HRL) is well suited to problems with a large state–action space. The core principle of HRL is to decompose a large problem into several smaller sub-problems based on a hierarchy. HRL algorithms can be implemented in various ways, including an Options Framework and Feudal RL.

The Options Framework consists of three components, namely a policy, a termination condition, and an initiation condition. The whole learning process of the Options Framework is modelled as a semi-MDP. At each stage, the agent at the higher level determines an action first. Then, the resulting state is compared with a list of initiation conditions. If an initiation condition is reached, then an agent at the lower level takes over the task. Similarly, the agent at the lower level hands over the task to the agent at the higher level once the termination criterion is reached (Bacon et al., 2017). Feudal RL is another type of HRL which imitates a feudal empire. At the higher level, decision-makers only need to consider their actions at their own level. The action made by the decision-makers is a goal for the sub-agents at the lower level, who attempt to fulfil the goal through a number of training steps. Afterwards, the high-level decision-maker would observe the new state and set a new goal. It is worth noting that the rewards and states are different at the two levels. Agents are allowed to retrieve only the information in their predefined granularity. The reward measure of a low-level agent is the fulfilment of the goal (Vezhnevets et al., 2017).

HRL has three key advantages. First, it improves exploration since the high-level agent can ignore implementation details. Second, for the same reason, agents have higher sampling efficiency. Third, since the low-level actions may share certain similarities, they can be transformed into different tasks in the same learning process. For example, if the high-level action is the assignment of destinations to vehicles and the low-level action is vehicle routing, the vehicle routing agent can be reused repeatedly for different assignments. However, it also exhibits common problems faced by other MARL algorithms, such as its limitations to solve problems which may not be decomposed into two-level decision problems and unstable performance.

2.4. Deep reinforcement learning

Standard RL frameworks may not perform effectively in many complex real-world systems with high-dimensional state and decision spaces (Haydari and Yilmaz, 2022). This challenge calls for recent research efforts in developing Deep Reinforcement Learning (DRL) algorithms, which adopt Deep Neural Networks (DNN) for approximating and learning the underlying value and policy functions. In addition to standard ANNs, various structures, including Recurrent Neural Network (RNN) and Graph Neural Network (GNN), are used as approximations of value functions and policy functions in optimal MDP with large state and decision spaces.

2.4.1. Deep Q network

Deep Q Network (DQN) is a basic DRL model in which the value or policy function is approximated by a multi-layer ANN, where an example can be referred to Fig. 1. The ANN could consist of multiple linear layers and element-wise non-linear activation functions (e.g., rectified linear unit (ReLU) and hyperbolic tangent (Tanh); see Goodfellow et al. (2016)). Mathematically, the ANN can be regarded as a generalised form of the linear kernel regression model adopted in various studies with the use of ADP (Yin et al., 2016; Liu et al., 2018; Ghasempour and Heydecker, 2020). The ANN covers the linear kernel regression model as a special (and simpler) case with a single layer of few elements, known as ‘neurons’ in ANN terminology (Bertsekas, 2019). With its sophisticated mathematical structure, the ANN can approximate and capture sophisticated system dynamics and stochasticity.

The training of the ANN adopts the concept of replay buffer (Haydari and Yilmaz, 2022; Ying et al., 2020), which is a data queue storing realised transition samples from successive stage pairs over a series of simulation runs. The replay buffer serves as a dynamic library of ground truths for training the ANN over epochs. With the sampled transitions in the replay buffer, the ANN can be trained by either a DQN algorithm (Mnih et al., 2015). The DQN training algorithm is a model-free off-policy TD learning approach. It is noted that the DRL process can be unstable when a highly nonlinear approximator (e.g., ANN herein) is used to represent the value and policy functions (Mnih et al., 2015). To ensure stability in the training process, target values will be used with which the ANN parameters are updated via a moving average approach (Lillicrap et al., 2015; Mnih et al., 2015).

The key ideas of the DQN algorithm are presented as follows. A batch of B transitions are sampled at each training epoch from the prevailing replay buffer, with which we can calculate the target value y_b of the Q -value realised from each of the sampled transitions b as:

$$y_b = r_b + \gamma Q_{\theta'}(s_{b+1}, \pi_{\phi'}(s_{b+1})),$$

$\forall b = 1, 2, \dots, B$ in the sampled batch; r_b is the reward gained in sample b ; $Q_{\theta'}$ and $\pi_{\phi'}$ are respectively the target critic and actor networks herein (Lillicrap et al., 2015; Mnih et al., 2015).

The ANN approximation Q_{θ} associated with its parameter set θ is updated based on the target values derived from the sampled batch. The concept herein aligns with the one-step TD learning (Sutton and Barto, 2018). The ANN parameter update is carried out via the following stochastic batch optimisation (Goodfellow et al., 2016), which seeks a target parameter θ' that minimises the loss function $\mathcal{L}_Q$ between each of the targets y_b and the corresponding Q -values in the batch:

$$\theta' = \arg \min_{\theta} \mathcal{L}_Q(\theta), \quad \text{where } \mathcal{L}_Q(\theta) = \frac{1}{B} \sum_{b=1}^B (Q_{\theta}(s_b, \mathbf{x}_b) - y_b)^2, \quad (9)$$

in which $\mathbf{x}_b = \pi_{\phi}(s_b)$ is the decision generated by the current policy network π_{ϕ} with the prevailing state s_b .

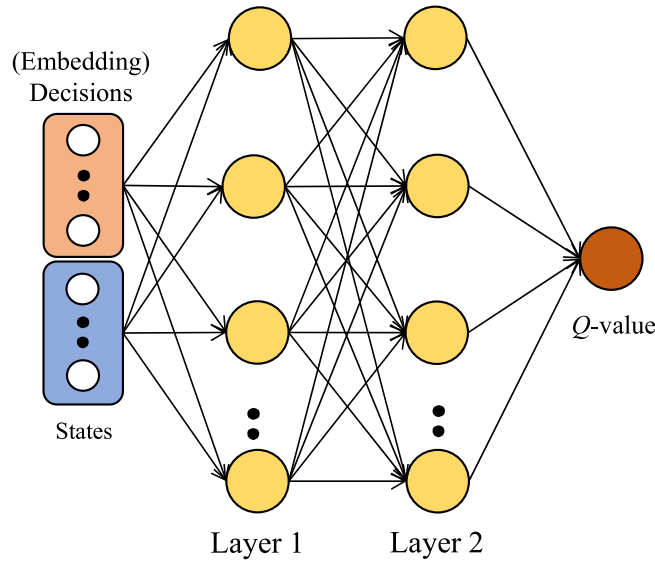


Fig. 1. An artificial neural network (Ying et al., 2020).

To ensure convergence and stability of the training process (Mnih et al., 2015), the target parameters in the ANN are updated via moving averaging:

$$\theta' \leftarrow \xi\theta + (1 - \xi)\theta'$$

where ξ is a sufficiently small smoothing factor.

The training process iterates for R times until the value converges, i.e., when there is no further possible change in the parameter set θ .

2.4.2. Actor-critic algorithms

Actor-critic (AC) algorithms combine uses of approximations for policy function and value-based function. In a DRL setting, this involves two multi-layer ANNs. In addition to the ANN shown in Fig. 1 for the use as a ‘critic’ function to approximate the value function, an AC framework includes an ‘actor’. This ‘actor’ works as a policy function approximation (as illustrated in Fig. 2), which derives the settings of the decisions based on the prevailing states and estimates by the critic network. Both actor and critic ANNs could consist of multiple linear layers and element-wise non-linear activation functions, e.g., ReLU and Tanh (Goodfellow et al., 2016).

The ANNs in the AC framework can be trained by a deep deterministic policy gradient (DDPG) algorithm (Lillicrap et al., 2015). Similar to DQN, the DDPG algorithm is a model-free off-policy training approach, which trains the AC agent from realised transitions. This approach does not require any prior knowledge of the underlying transition model. The training process starts with setting up a replay buffer storing realised transition samples.

In each training epoch, a decision is first determined by the actor ANN, based on the prevailing state and its parameters settings. To avoid being trapped with an inefficient decision policy, Gaussian noise is associated with the decision derived in the process. The noise follows a Gaussian distribution with a mean of zero and a finite standard deviation ϵ^r depending on the training epoch r , as presented in Eq. (10).

$$\epsilon^r = \epsilon^{\min} + \frac{1}{2}(\epsilon^{\max} - \epsilon^{\min}) \left[1 + \cos \left(\min \left\{ \frac{r}{R_e}, 1 \right\} \pi \right) \right], \quad (10)$$

where $\epsilon^{\min}$ and $\epsilon^{\max}$ are respectively the lower and upper bounds set on ϵ^r , R_e is the epoch at which the noise to prevent the process from further decaying, and π refers to pi radians in this context. ϵ^r is known as the explorative noise, and its standard deviation is set to be decreasing over the training epoch r for convergence (Lillicrap et al., 2015). This noise component helps balance exploration and exploitation in the training process (Powell, 2011; Bertsekas, 2019). An example can be referred to Ying et al. (2020).

With both value function (critic) and policy function (actor) being approximated by ANNs, we have a training process associated with the actor ANN (say, with parameters π_ϕ) in addition to the value function approximation update presented in Section 2.4.1.

Again, given a batch of B transitions sampled from the current replay buffer, with which we will calculate the target value y_b for updating the value function as discussed previously, we also have the actor ANN to be updated by seeking a target parameter set ϕ' for it that minimises the expected Q -value generated by the updated critic ANN over all state-decision pairs (s_b, x_b) in the sampled batch as:

$$\phi' = \arg \min_{\phi} \mathcal{L}_A(\phi), \quad \text{where } \mathcal{L}_A(\phi) = \frac{1}{B} \sum_{b=1}^B Q_{\theta}(s_b, \pi_{\phi}(s_b)). \quad (11)$$

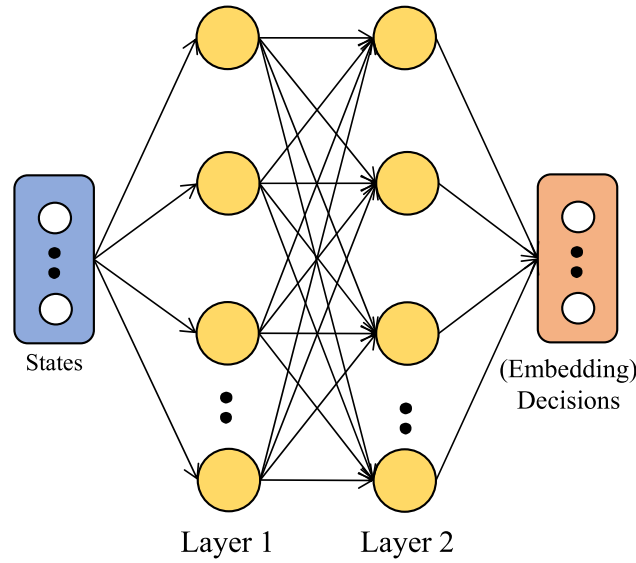


Fig. 2. An example of an ANN-based policy approximation (Ying et al., 2020).

The Minimisation Problem (11) can be solved by a stochastic gradient search with the gradient $\nabla_{\phi} \mathcal{L}_A(\phi)$ conditioned on ϕ . The gradient can be derived following the chain rule and the deterministic PG theorem proposed by Silva et al. (2019) as

$$\nabla_{\phi} \mathcal{L}_A(\phi) = \mathbb{E} \left[\nabla_{\bar{x}_b} Q_{\theta}(s_b, \bar{x}_b) |_{\bar{x}_b = \pi_{\phi}(s_b)} \nabla_{\phi} \pi_{\phi}(s_b) \right]$$

in which the terms on right hand side computes expectation over the multiplication of the gradient of critic ANN on the decision $\mathbf{x}_b$ (which equals $\pi_{\phi}(s_b)$) and the gradient of actor ANN π_{ϕ} . There are effective algorithms available for solving Optimisation Problems (9) and (11), such as the well-established Adam algorithm (Kingma and Ba, 2014).

To ensure convergence and stability of the training process (Mnih et al., 2015), the target parameters in the actor ANN are updated via the moving averaging:

$$\phi' \leftarrow \xi \phi + (1 - \xi) \phi'$$

where ξ is a sufficiently small smoothing factor.

The training process iterates for R times. With the decaying explorative noise ϵ^r over training epochs r , the training process gradually converges to a stable solution.

2.4.3. Recurrent neural network

The underlying ANNs in a DRL framework could be replaced with some other more advanced architectures. Recent advances in neural language processing show the strength of Recurrent Neural Networks (RNNs). Instead of having a fixed number of inputs and outputs as in standard ANNs, RNNs can incorporate any arbitrary length. Fig. 3 provides an example of an RNN (Goodfellow et al., 2016). In this example, an input $\mathbf{x}$ is transformed to an output $\mathbf{o}$, through a series of transformations governed by the loss function L and other parameters determined in the training process.

As an extension of RNN, Long Short Term Memory (LSTM) (Hochreiter and Schmidhuber, 1997) has been widely applied in various areas focusing on sequential data-related problems. LSTM has a gated structure that selectively magnifies or weakens the hidden state information. An LSTM cell is a basic component in the LSTM module. It is constructed with three gates, including the input gate i_t , the forget gate f_t , and the output gate o_t , which control the information flow over time step t as shown in Fig. 4.

We further denote $\mathbf{x}_t$ and $\mathbf{h}_t$ the input and hidden state of the LSTM cell at time step t respectively herein. A memory cell state $\mathbf{C}_t$ is utilised to record the current state of the LSTM cell. The calculation of $\mathbf{C}_t$ and $\mathbf{h}_t$ are formulated as the following steps. First, the three types of LSTM gates are computed via the system of Eqs. (12):

$$\begin{aligned} i_t &= \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1}) \\ f_t &= \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1}) \\ o_t &= \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1}) \end{aligned} \quad (12)$$

where $\sigma(\alpha)$ denotes a sigmoid function (which performs element-wise operations if a vector is an input):

$$\frac{\exp(\alpha)}{\exp(\alpha) + 1};$$

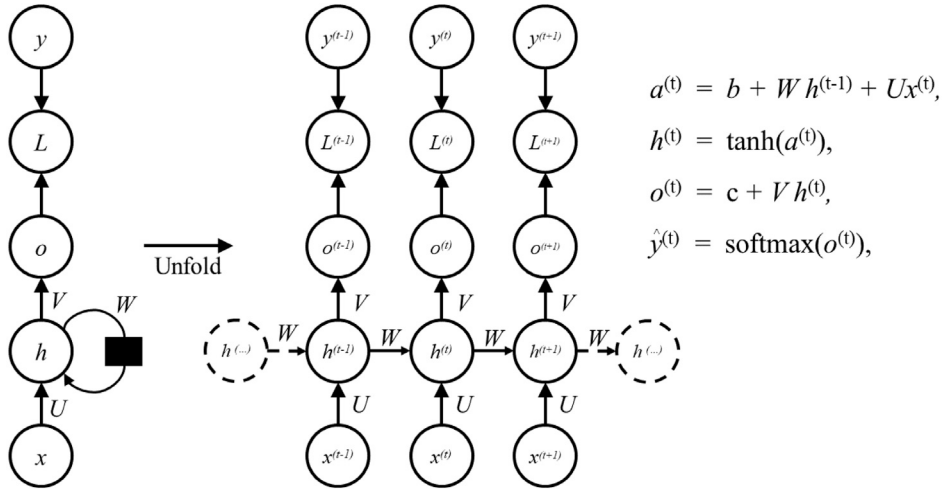


Fig. 3. An example of an RNN (Goodfellow et al., 2016). In this example, h denotes the hidden state variables in the RNN, while V , W , and U are the underlying learnable parameters. The parameters c and b refer to the bias terms. o is the output from the model, x denotes the input data, and y denotes the actual observations. Function L is the loss function for the training process, while the superscript t denotes the (time) index of a cell (within the NN).

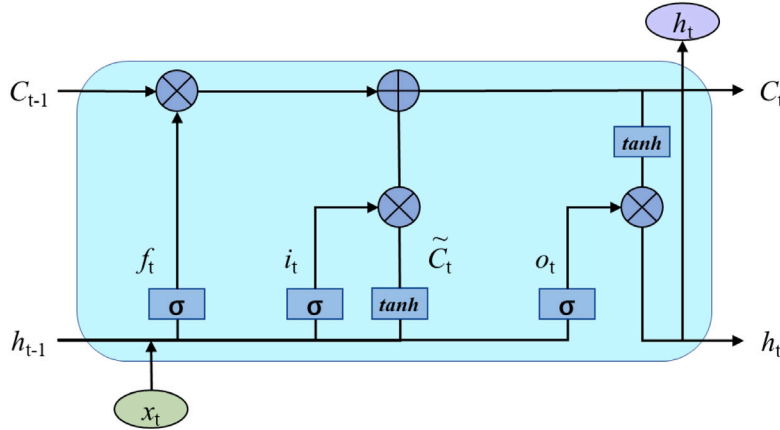


Fig. 4. An illustration of an LSTM cell (Hochreiter and Schmidhuber, 1997).

h_{t-1} represents the previous hidden state, i.e., at time step $t - 1$; W_i , W_f , and W_o , as well as U_i , U_f , and U_o , are the weight matrices that transform the values of x_t and h_{t-1} to the input, forget, and output gates, respectively.

The cell input $\tilde{C}_t$ is calculated in Eq. (13), while the cell state C_t is obtained according to Eq. (14).

$$\tilde{C}_t = \tanh(W_g x_t + U_g h_{t-1}) \quad (13)$$

$$C_t = i_t \circ \tilde{C}_t + f_t \circ C_{t-1} \quad (14)$$

where $\tanh(\alpha)$ denotes a tanh function (which performs element-wise operations if a vector is an input):

$$\frac{\exp(\alpha) - \exp(-\alpha)}{\exp(\alpha) + \exp(-\alpha)};$$

W_g and U_g represent the weight matrices that transform the values of x_t and h_{t-1} to the cell input; $\circ$ denotes the Hadamard product of two matrices; and C_{t-1} is the previous state of the cell, i.e., at time step $t - 1$.

Finally, the hidden state of the LSTM cell can be determined from Eq. (15):

$$h_t = o_t \circ \tanh(C_t) \quad (15)$$

In the LSTM module, a series of LSTM cells are connected sequentially over time. Temporal features in historical data are passed through the module and encoded in the hidden state of the LSTM cell h_t .

For problems with finite action spaces, pointer network and attention mechanism, proposed by Vinyals et al. (2015), can be applied. A pointer network gives softmax encoded solutions, each of which reflects the probability of selecting a solution. In an

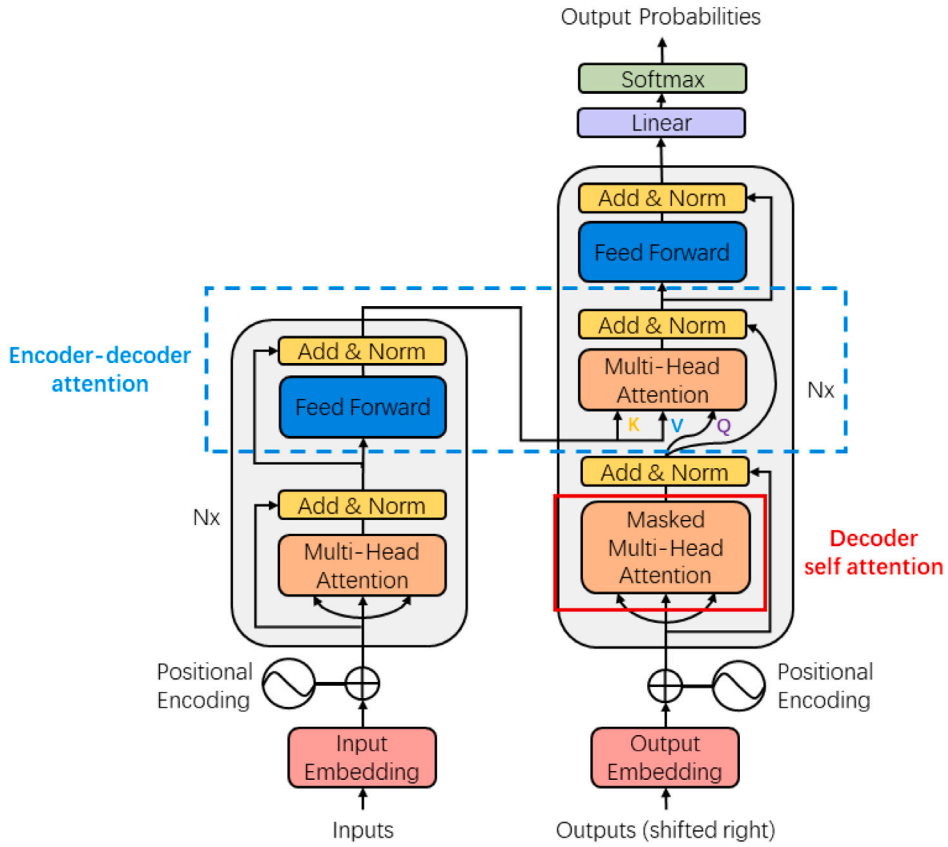


Fig. 5. An example of a transformer architecture (Vaswani et al., 2017).

example of finding a convex hull of points, the inputs to the encoder are the coordinates of each point, and the output of each encoder cell is softmax encoded with the probabilities of having any set of coordinates in a sequence. In this example, suppose there are four nodes, then each decoder has five options, with the last output corresponding to the termination option.

Traditional RNN techniques pass information to downstream cells recursively. It may suffer from computational instability, such as diverging and undefined gradients. Vaswani et al. (2017) propose a novel integrated attention-RNN method, termed transformer architecture. An example of a transformer architecture is depicted in Fig. 5. Instead of recursively passing hidden states to downstream cells, this architecture directly computes the embedding of each input element and then gather the information through a multi-head attention mechanism.

2.4.4. Graph neural network

In addition to RNN and LSTM, Graph Neural Network (GNN) has been gaining popularity in RL-based optimisation. Fig. 6 shows an example of a GNN architecture, and a comprehensive survey on GNN can be referred to Peng et al. (2021).

There are numerous GNN variants, which include Graph Convolutional Network (GCN), Graph Attention Network (GAN), Graph Isomorphism Network, and Structure-to-Vector Network. Typically, a GCN (Kipf and Welling, 2017) utilises the following mathematical formulation:

$$h_{i(l+1)} = \sigma \left(\mathbf{W}_{l+1} h_{il} + \sum_{j \in N(i)} \frac{1}{c_{ij}} \mathbf{U}_{l+1} h_{jl} \right)$$

where $\mathbf{W}_l$ and $\mathbf{U}_l$ are weight matrices for nodes at layer l , $N(i)$ is the set of nodes neighbouring to node i , c_{ij} is the normalisation constant for the pair of nodes (i, j) , h_{il} is the nodal embedding i at layer l . The intuition of this formulation is to exchange information between nodes, so that the embedding of each node can carry information not only from itself, but also the entire graph. The embeddings are passed onto decoders customised for individual problems for further processing.

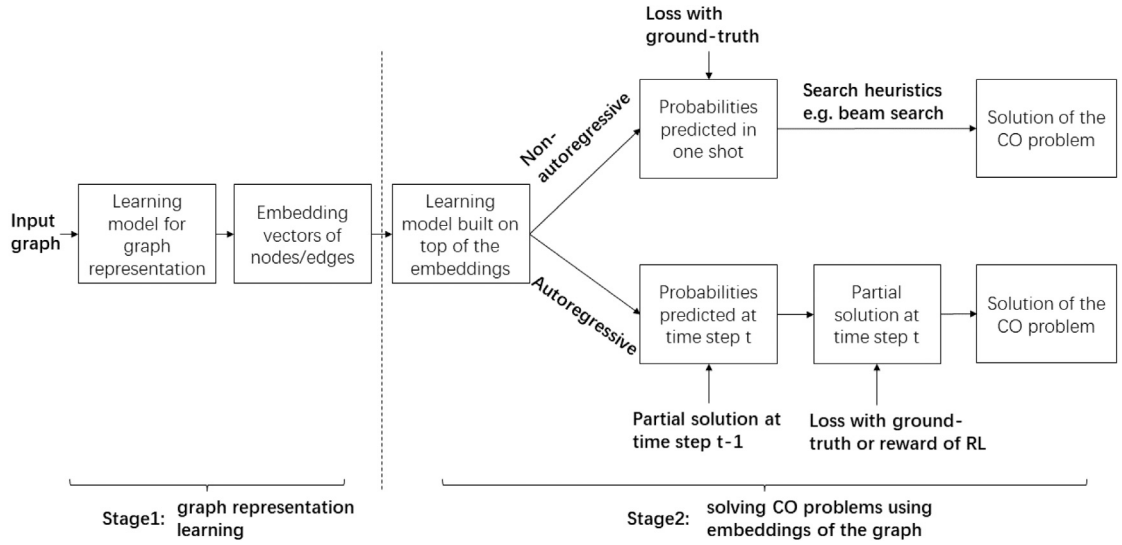


Fig. 6. An example of a GNN architecture (Peng et al., 2021).

On the other hand, GAN is an extension of GNN (Veličković et al., 2018). This percentage output of the softmax function α_{ij} reflects the importance of the information from other nodes to the current node:

$$h_{i(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij} \mathbf{W}_l h_{jl} \right)$$

$$\text{where } \alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}_l^T [\mathbf{W}_l h_{il} \parallel \mathbf{W}_l h_{jl}]))}{\sum_{k \in N(i)} \exp(\text{LeakyReLU}(\mathbf{a}_l^T [\mathbf{W}_l h_{il} \parallel \mathbf{W}_l h_{kl}]))},$$

in which the notation T denotes the transpose of the associated matrix; $\parallel$ denotes the concatenation operation of tensors; and *LeakyReLU* is a rectifier or activation function that is commonly used in an RL framework. Given an input x , the *LeakyReLU* function is defined as

$$\text{LeakyReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ 0.01x & \text{otherwise} \end{cases}$$

The *LeakyReLU* function enables the neurons in the ANN to capture non-linearity in the input data.

2.4.5. Convolutional neural network

Similar to multi-layer perceptrons, Convolutional Neural Network (CNN) is a black box model consisting of numerous parameters to determine. The main difference between these two approaches is that multi-layer perceptrons are fully connected, while CNN is partially connected. In transportation and logistics applications, CNN has been applied for image processing. For example, images captured from maps or vehicle cameras can facilitate routing decisions. CNN requires several processing steps to extra features from an image, including convolution, padding, pooling, and flattening. Convolution, the first processing step, involves a kernel for initial feature extraction. The kernel consists of parameters and a bias term that are shared across the entire image. The convolution process begins from the upper left corner of the picture. The inputs are multiplied by the respective kernel parameters to obtain a value, which corresponds to the convolved feature of this region. The kernel then moves for one stride step to obtain the feature of the following region until a matrix of convolved features is obtained. An image may undergo several convolution processes to construct the final feature matrix. During this process, since each region of the image corresponds to a value, the dimension of the matrix would be reduced after each convolution. Padding is an operation to add zeros around the boundaries of the convolved features, such that the dimension can be consistent throughout the entire process. Pooling is an operation to reduce the computational requirement by extracting only the critical features from the feature matrix. There are two popular pooling approaches: max pooling and average pooling. Max (and average) pooling utilises the maximum (and average, respectively) value of a region in the convolved feature matrix. Max pooling usually works better than average pooling because it also suppresses noises. After pooling, another cycle of convolution, padding, and pooling follows, until the preselected number of cycles is reached. The final feature matrix is converted to a high dimensional array representation of the image through fully connected multi-layer perceptrons. This high dimensional matrix can then be used to conduct further operations such as image classification or labelling. In a logistics and SC context, this high dimensional array can be used to map to specific actions or reward predictions.

2.4.6. Fuzzy Inference

Fuzzy inference is a process that maps inputs to outputs, through a fuzzifier, a fuzzy logic processor, and a defuzzifier. At the first step, the fuzzifier maps crisp inputs into fuzzy values through membership functions. Such a conversion process avoids black-and-white discriminative rules, and allows the values to be defined in a probabilistic sense. The fuzzy logic processor requires a set of rules to map the fuzzified inputs to outputs. This fuzzy logic processor then outputs fuzzy probabilities according to the specific rules. At the defuzzification step, the fuzzy probabilities are mapped to exact control actions.

Fuzzy inference is a mid-course between system models and fully parametric models. It requires less domain knowledge compared to a system model. In many applications, the full system is often difficult to be formulated mathematically. With fuzzy inference, one can train the parameters of the model with real data, similar to the way of training a NN. However, NN is a flexible black box model, where prior domain knowledge is not required. On the other hand, fuzzy inference requires assumptions on the membership function and certain knowledge about the fuzzy logic (in the form of “if-then” rules), which can be difficult to define. Despite the disadvantage of having assumptions, fuzzy inference enables faster training and better explainability due to its significantly fewer parameters (Kumari and Sunita, 2013).

2.5. Hybrid reinforcement learning and optimisation

The previous RL frameworks can be regarded as pure data-driven or model-free solution approaches, because they merely rely on observations/realisations of the actual states and any optimisation model. A pure data-driven system would need to be trained with a significant number of observations/realisations for satisfactory performance. To improve the overall system performance, one could adopt a hybrid RL and optimisation approach. In a hybrid framework, the decision $\mathbf{a}_{t_0}$ at the current stage t_0 is derived from the following optimality condition:

$$\mathbf{a}_{t_0}^* = \arg \max_{\mathbf{a}_t} \mathbb{E}_{\pi} \left[\sum_{t=t_0}^{t_0+T-1} \gamma^{t-t_0} \hat{r}_t + \gamma^T \tilde{V}_{\pi}(\hat{\mathbf{s}}_{t_0+T}) \right], \quad (16)$$

where T is the length of the horizon specified in the optimisation model and $\hat{r}_t$ is the estimated reward at stage t . With the use of the optimisation model, the data-driven value function approximation $\tilde{V}_{\pi}$ in Eq. (16) is used to predict the reward beyond the modelling horizon T from the current stage t_0 , based on the estimated state $\hat{\mathbf{s}}_{t_0+T}$ right after the end of the decision horizon. With the value of $\mathbf{a}_{t_0}^*$ from Eq. (16) at stage t_0 , the value function approximation $\tilde{V}_{\pi}$ is updated for the next decision stage $t_0 + 1$ through TD learning.

2.6. Bayesian reinforcement learning

A challenge in RL lies in balancing the trade-off between exploration and exploitation. More specifically, the performance of an RL approach highly depends on the amount of effort to maintain the solution quality (exploitation) and the time to explore unknown solution space (exploration). The trade-off between exploration and exploitation is a concern in both model-based and model-free approaches. To this end, Bayesian RL provides a systematic approach to addressing such a trade-off. By providing the prior information about the uncertainties of the RL system (such as transition probabilities and rewards), Bayesian RL applies Bayesian inference to update the prior belief of uncertainties after each transition. This posterior distribution following the Bayesian update, combined with the physical state defined in the problem, is called hyper-state. With consideration of transitions among different hyper-states, the resulting RL models consider the exploration-exploitation trade-off as part of the decision-making problems. In what follows, we will provide a brief introduction to Bayesian RL, and we refer the reader to Ghavamzadeh et al. (2015) for the details.

The main idea of Bayesian RL is to incorporate the belief of the uncertainties into the state space in the MDP. In particular, we consider the hyper-state (s, ϕ) , where ϕ is the distribution of the uncertain parameters. To illustrate the main idea, we simplify our notation by considering that the reward function is known, i.e., $r_{s,a} = r_{s,\phi,a}$. Note that this setting of Bayesian RL can be generalised to the cases where the reward function is unknown. We refer to this model as Bayes-Adaptive MDP.

In the above setting, the model considers the hyper-states with the transition function

$$Pr(s', \phi' | s, a, \phi) = Pr(s' | s, a, \phi) Pr(\phi' | s, a, s', \phi),$$

where both $Pr(s' | s, a, \phi)$ and $Pr(\phi' | s, a, s', \phi)$ are known; the first term is known because the transition probabilities are given in ϕ , and the second term can be computed via Bayesian update (where $Pr(\phi' | s, a, s', \phi) = 1$ if ϕ' is the posterior distribution given the prior distribution ϕ and sample (s, a, s') , and $Pr(\phi' | s, a, s', \phi) = 0$ otherwise).

Thus, one could derive an optimal strategy that automatically balance the exploration-exploitation trade-off by solving the above Bayes-Adaptive MDP. Despite its advantages, the computational cost of solving Bayes-Adaptive MDPs is much higher, compared to nominal MDPs. In particular, even if the physical state space is finite, the state space of the corresponding hyper-state could be infinite, as ϕ is a probability distribution. Moreover, the Bayesian update that is used to compute $Pr(\phi' | s, a, s', \phi)$ is computational intractable in general. One can overcome these computational challenges by carefully selecting the prior distribution. For example, suppose the transition probabilities $p_{s,a}$ at state s with action a follows a Multinomial distribution with unknown parameter $p = (p_{s,a,1}, p_{s,a,2}, \dots)$. Then, by assuming that the prior distribution of p is a Dirichlet distribution with parameter $\phi = (\phi_{s,a,1}, \phi_{s,a,2}, \dots)$, the posterior distribution after observation s' is also a Dirichlet distribution with updated parameter ϕ' , where $\phi'_{s,a,s''} = \phi_{s,a,s''} + 1$ if

$s'' = s'$ and $\phi'_{s,a,s''} = \phi_{s,a,s''}$ otherwise. Thus, analytical Bayesian update is available in this case, which is computationally attractive. Yet, the computational cost of solving the associate Bayes-Adaptive MDP would still be extremely expensive, as the state space of the Bayes-Adaptive MDP is $O(|S|^T)$ where S is the original physical state space and T is the horizon of the problem; this is because more Bayesian updates are performed as for larger T and allow ϕ to deviate more from the initial ϕ_0 , and so the volume of the state space of ϕ increases.

Since Bayes-Adaptive MDPs are difficult to solve in general, various algorithms have been developed to approximate the optimal solution. Many of them are adopted and modified from existing solution methods for solving MDPs, partially observable Markov decision processes (POMDPs), and stochastic bandits. For example, both finite-state controllers and Bayesian Exploration Exploitation Tradeoff in Learning are offline algorithms that obtain tractability via value approximation. Bayesian DP, forward search planning, and Bayes-adaptive Monte-Carlo planning are other examples of computational methods that solve Bayes-Adaptive MDPs.

On the other hand, Bayesian inference can also be applied in a model-free setting, in which value function or policy are updated via new sample transitions. In particular, Gaussian Process Temporal Difference Learning (GPTD) applies Gaussian Processes to evaluate value function for any given policy. Building upon the same idea as GPTD, GPSARSA estimates the Q function using Gaussian Processes and determines the action at each time step via sampling and estimating the Q function over action space. Other than methods that are based on value function, Bayesian PG, and Bayesian AC adopt the ideas from PG methods in nominal RL settings.

3. Applications

In this section, we conduct a review of the applications of RL for logistics and SCM.

3.1. Article selection methodology

Following some recent literature reviews in transport and logistics (e.g., Cheung et al., 2021; Cavallaro and Nocera, 2021), we first present the article selection methodology adopted in our review.

3.1.1. Synthesis method

We adopt a scoping review approach (Pham et al., 2014), which is a class of emerging synthesis methods, to conduct our literature review. A scoping review approach maps the existing literature on a particular research topic for the purpose of identifying key concepts, theories, sources of evidence, and research gaps (Grimshaw, 2010; Levac et al., 2010; Pham et al., 2014). It is different from systematic reviews, which intend to examine the appropriateness and effectiveness of methodologies (Munn et al., 2018). We adopt this review approach for the following reasons. First, there are different types RL applications in the domain of logistics and SCM. The research objectives and problem definitions of these studies have a wide spectrum. Second, because of the wide range of applications, numerous RL approaches have been customised to solve specific problems. Third, the applications are typically highly complicated and of large scale such that the RL approaches may not provide theoretical performance guarantees. For performance evaluations, multiple branches of optimisation approaches (e.g., exact algorithms, heuristics, and existing RL methods) have been used to provide benchmarks. Last but not least, a significant number of research studies are published in different categories of venues (e.g., journals, conference proceedings, book chapters, and archives) due to the cross-disciplinary nature of the topic and industry-oriented applications. For the above reasons, we adopt a scoping review approach such that this great variety of studies can be categorised into subtopics to summarise their problem natures and RL methodologies developed, document the evidences of applications, and identify the research gaps.

3.1.2. Search terms

We first conduct a literature search on *Scopus* for publications using the query syntax: TITLE-ABS-KEY("reinforcement learning") AND TITLE-ABS-KEY("logistics" OR "supply chains"). With the publications identified in the first literature search, we read the titles, abstracts, and keywords of the publications and identify further keywords that are relevant to this survey. A second literature search based on the following query syntax is conducted using *Scopus* to identify more relevant studies: TITLE-ABS-KEY("reinforcement learning") AND TITLE-ABS-KEY("air" OR "aviation" OR "delivery" OR "demand prediction" OR "drone" OR "facility" OR "inventory" OR "logistics" OR "port" OR "pricing" OR "railway" OR "repositioning" OR "resource allocation" OR "timetabling" OR "unmanned aerial vehicle" OR "UAV" OR "vehicle routing" OR "VRP"). We do not restrict the publication period in our search.

3.1.3. Selection, inclusion, and exclusion criteria

To identify the most relevant publications, we consider those published in the journals under the three categories of "Operations Research & Management Science", "Transportation", and "Transportation Science and Technology" in *Journal Citation Reports*. The initial selection of papers (conducted on June 7, 2021) results in 320 journal articles. We go through these publications and exclude those not within the scope of this review. According to our scope, traffic engineering (including signal control and autonomous driving), electric engineering, transport of passengers, retailing, and manufacturing processes are excluded. After those papers are excluded, 72 journal articles remain for our comprehensive review. Since RL is a key area of computer science and an emerging area in other fields, we also include other significant recent research studies available in *arXiv*, conference proceedings (e.g., AAAI,

HICSS, ICLR, ICML, and *NeurIPS*), and other journals. The authors also include further research articles based on their knowledge to strengthen the comprehensiveness of this literature review. Furthermore, new papers published during the revision of our study are included. There are a total of 136 research studies reviewed in this paper. The detailed statistics of the reviewed studies will be presented in Section 4.1.

We note that there could be limitations in the article selection methodology. First, publications not written in English and those which are not available in the *Scopus* database may not be identified using the search terms. Second, some authors did not label the paper with the keyword “reinforcement learning”, but with specific algorithms such as “DDPG” or “SMART”. Likewise, specific logistics and SCM applications may use variants of our keywords. To overcome these limitations, our article selection methodology involves a step of identifying papers based on authors’ knowledge and tracing articles (which cite or are cited by the identified research studies) to enhance the comprehensiveness of the pool of literature. This step is necessary, in particular for those computer science research studies, which are often made available before publication.

3.2. Organisation of the review of applications

We review and classify applications of RL into two categories: *logistics* and *SCM*. Under logistics, the applications are related to the detailed operational plans to guide the movements of products and goods. Under *SCM*, the applications include other relevant processes to enable the delivery of products and goods from the beginning of the SC to the final consumers.

The reviewed articles can be characterised by three dimensions: application domains, levels of decisions (if possible), and methodologies. Articles are first categorised by their applications and then levels of decisions (if any), where methodologies are discussed. Characteristics of the reviewed articles under the same application are summarised at the end of the subsection.

3.3. Logistics

Under logistics, we further categorise research studies according to the application domain.

3.3.1. Travelling salesman problem and vehicle routing problem

A popular research area that aims to assess the performance of RL approaches is to develop highly efficient RL algorithms to solve hard combinatorial optimisation problems. NP-hard problems, such as the knapsack problem, TSP, and VRP, which are typically solved by exact algorithms or heuristics, are recently tackled in the RL community. Wang and Tang (2021) provides a comprehensive survey of the studies on RL for combinatorial optimisation. This section focuses on those combinatorial optimisation problems related to logistics, where most of which are related to TSP and VRP. We do not intend to review all the studies related to TSP and VRP. Our focuses in this subsection are to discuss RL studies which are more relevant to logistics and identify gaps between the use of existing algorithms and real applications. We refer the reader to other reviews for comparison of algorithmic efficiency, computational complexity, and result quality. Farazi et al. (2021) reviewed RL in transport and traffic domain; Mazyavkina et al. (2021) conducted a survey that covers RL combinatorial optimisation; demand driven transport network optimisation is surveyed by Zong et al. (2021)

Bello et al. (2016) propose an encoder–decoder gated RNN algorithm to solve TSP. For TSP, they randomly generate maps of squares and input the Euclidean coordinates of the nodes into the encoder to train the parameters. On the decoder side, the sequence of nodes to be visited is exported with pointer probabilities. The training is done with both RL and supervised learning approaches. Their results suggest that the RL training approach enables the algorithm to find more effective solutions more efficiently. Nazari et al. (2018) investigate the use of LSTM-based RNN for VRP. They argue that LSTM is designed originally for a machine translation setting with a strong coupling between contexts, whereas the order of the input coordinates does not matter. Therefore, they propose a simple architecture consisting of an embedding layer, a self-attention layer, and a pointer network. The authors find that this algorithm outperforms Google’s OR-tools in more than 60% of the instances. Kool et al. (2019) solve a number of routing problems, including TSP, CVRP, VRP with split deliveries, and orienteering problem, with a transformer architecture. Their results suggest that their proposed method based on attention layers can beat heuristics as well as LSTM in terms of both solution quality and computational efficiency.

Some studies consider VRP under uncertainty and model state transition. For example, Joe and Lau (2020) provide an algorithm for dynamic VRP with stochastic customer arrivals. One may refer to the work of Hildebrandt et al. (2021), who have provided a comprehensive review of dynamic VRPs studied in both the operations research and computer science communities.

There are other studies which conduct research on this topic with similar approaches; the embedding of the nodes are first derived and then multiplied to obtain the probabilities of transiting from one node to another. However, the multiplication of embeddings results in a large number of possibilities, leading to combinatorial search. Most of the studies simply apply beam search or greedy search approaches, without allowing a possibility of improvement. Hottung et al. (2020) propose a learning-based variational auto-encoder, which allows a guided search to improve the solution quality of the RL outputs, though a higher computational cost is required. It offers the users the flexibility to manage the trade-off between computational costs and effectiveness.

There are studies that propose combinations of methodologies, for example, combining heuristics or other predefined algorithms as pre-processing and/or post-processing tools or actions (Kalakanti et al., 2019; Zhao et al., 2020). Kalakanti et al. (2019) apply geometric clustering before solving VRP with a QL solver. Lu et al. (2019a) propose a learning improvement heuristic, which is an RL-based heuristic. They adopt GAN as the agent to learn from the state, including the available capacities of the vehicles and

Table 2
Summary of applications for TSPs and VRPs.

Study	RL agent	Training method	Problem
Bello et al. (2016)	RNN	PG	TSP, Knapsack
Nazari et al. (2018)	Self-attention	PG	TSP
Chen and Tian (2019)	DNN	AC	CVRP
Kalakanti et al. (2019)	Transformer	PG	VRP
Kool et al. (2019)	Transformer	PG	TSP, VRP
Lu et al. (2019a)	DNN	PG	VRP
Drori et al. (2020)	GNN	PG	TSP
Joe and Lau (2020)	DNN	DQN	Dynamic VRP
Zhao et al. (2020)	Transformer	PG	VRP
Zhang et al. (2021)	Transformer	PG	Dynamic TSP

distances between customers, and then choose actions from two different groups of heuristic operators to improve the solution quality until convergence. [Chen and Tian \(2019\)](#) train RNN for solving VRP with an improvement heuristic. The RNN encodes the current solutions and the map, and then determines the improvement rules and nodes.

[Table 2](#) summarises the RL applications for TSPs and VRPs. While the research on RL for solving TSP and VRP is fruitful, there are some research gaps to be filled. First, to the best of our knowledge, the majority of the RL algorithms developed for TSP and VRP use Euclidean coordinates as input, but do not have the capability of solving problems with general distance metrics. In practice, travel times depend on various factors, including road network distances and traffic conditions. New methodologies are required to train RL algorithms from generic origin–destination matrices for practical TSP and VRP. Recently, [Zhang et al. \(2021\)](#) attempts to apply GNN to solve a TSP based on varying traffic conditions. The research can be further extended to other relevant VRPs. Second, most of the studies train the parameters and solve the problem with identical sizes, i.e., the numbers of nodes of the training and test instances are the same. The performance analysis of the RL model for solving problems of scales larger than the training instances is still lacking. Third, computational instances are mostly randomly generated on maps of squares. Little research has been conducted on maps with highly skewed coordinates, which are common in practice.

3.3.2. Air logistics

We find that the applications of RL for air logistics are heavily inclined to the deployment of unmanned aerial vehicles (UAVs). Benefited from the recent technological development, UAVs or drones offer an alternative channel to our modern intelligent logistics systems. Given its potential positive impact on society, researchers have been investigating how UAVs could be optimally used by and complement the existing logistics systems. We classify these studies into two categories: assignment and routing problems.

In the assignment problems, decision-makers allocate UAVs to complete tasks. [Chen et al. \(2021c\)](#) study a same-day delivery problem, where the decision-maker allocates vehicles and drones to deliver items to customers. They model this problem as an MDP. At each stage, a service request is received. An RL agent decides whether to accept this request and updates the route plan if needed. By using DNNs to approximate the Q -function, this study proposes deep QL to solve this problem. [Khamidehi et al. \(2021\)](#) study a dynamic UAV assignment problem, where a number of UAVs are distributed to multiple package distribution centres and deliver goods in the assigned region. There are RL agents and a central scheduler in their RL framework. Each package distribution centre is represented by one RL agent, which computes the number of UAVs needed for a centre so that the probability of having delays is low. The central scheduler then assigns the UAVs to the centres based on the suggestions from the RL agents. Each RL agent considers the queue of jobs at the centre, and a double deep Q -Network solves this proposed RL model. [Asadi and Pinkley \(2021\)](#) propose an RL method for medical equipment delivery, where there are a depot and multiple locations with stochastic demands. Other than routing problems, an RL agent determines how much battery to charge for each drone.

Much research has considered the problems of determining optimal routes under different scenarios. [Faust et al. \(2017\)](#) study a cargo delivery problem, in which a UAV delivers an item from one place to another, while reducing unnecessary swings as well as avoiding obstacles in its path. A deterministic MDP in a continuous state space with finite actions is proposed to determine a minimal oscillation policy. To enhance tractability, a linear function application is applied to approximate the value function of the proposed MDP. [Feng \(2020\)](#) studies a similar problem where an RL model is proposed for the routing of multiple homogeneous UAVs. Their problem is solved by a PG method. The agent is a standard embedding plus an attention mechanism architecture. This architecture embeds current load, location, and task information and then chooses the next node to visit at each decoding step. [Wu et al. \(2021\)](#) study a truck and drone coordinated delivery problem. At the initial step, a k-means clustering algorithm is used to find the optimum parking location of the truck. A transformer architecture is constructed to derive the sequence of drone visits at all locations. [Alkouss and Bouguettaya \(2021\)](#) propose a congestion aware QL algorithm for drone swarm services. The RL agent determines the sequence and times of visits.

On top of standard routing problems, some researchers have investigated how UAVs could be integrated with other infrastructures. For example, [Samir et al. \(2020\)](#) consider a UAV-assisted single-hop vehicular network where UAVs receive data from different vehicles. The goal is to minimise the age of information by optimising paths and the scheduling policies of UAVs. A DRL framework is proposed, where the (action-)value function approximation is obtained by DNNs. Since the action space is continuous, this problem is solved using an AC algorithm, called DDPG. [Zeng and Xu \(2019\)](#) study how UAVs could be benefited by existing communication networks, where UAVs are considered to be aerial user equipment with different missions in a cellular network. To

Table 3
Summary of applications for air logistics.

Study	Method(s)	State	action	Problem
Faust et al. (2017)	Approximate value iteration	UAV positions, velocities, and orientations	UAV accelerations	Path planning
Zeng and Xu (2019)	$Q(\lambda)$ -learning	UAV positions	UAV travelling directions	Path planning
Feng (2020)	Attention + PG	Customer orders and their locations	Routes	VRP
Samir et al. (2020)	DDPG	Traffic conditions, UAV positions, and age of information	UAV assignments, travelling distances and directions	Path planning and scheduling
Alkouz and Bouguettaya (2021)	DQN	Unclear	Time slots of visits	Drone reallocation
Asadi and Pinkley (2021)	Tabular QL	Battery levels	Recharging level	Assignment and battery charging
Chen et al. (2021c)	DQN	Time, customer locations, and planned routes	Acceptance decisions and new routes	VRP
Khamidehi et al. (2021)	Double DQN	Number of UAVs and the queue length in the package distribution centre	Re-distribution of UAVs among package distribution centres	Assignment
Wu et al. (2021)	Transformer	Customer locations	Sequence of drone visits	Delivery by trucks and drones

maintain high connectivity in the network, the authors propose an RL approach to compute communication-aware paths, in which the optimal paths minimises the travel time while receiving sufficient signal. The proposed RL model is then trained by TD learning. Linear approximation is used to approximate the value function as the state space is continuous, and the associated feature vector is constructed via tile coding.

Table 3 summarises the characteristics of the studies of air logistics. Given the unique features of UAVs, it would be interesting and important to conduct further research on (1) three-dimensional UAV routing problems (as UAVs may need to fly at different heights in urban areas subject to physical and governmental restrictions), (2) stochastic UAV routing problems (because UAV operations can be affected by environmental conditions such as wind, signal communications, and obstacles) (3) enhanced integrated operations between UAVs and the other essential resources (e.g., trucks, vessels, and human couriers) in modern logistics systems.

3.3.3. Maritime logistics

The applications of RL on maritime logistics include yard cranes' sequencing, yard truck deployment, ship block arrangement at stockyard, as well as vessel path planning. The RL techniques span a variety of methods, such as QL and Asynchronous advantage actor-critic (A3C) algorithm. The details are analysed in this subsection.

At a maritime container terminal, vessels deliver containers to the port, where quay cranes can retrieve containers. Then, the retrieved containers will be loaded onto yard trailers (or trucks) that move around the port area. To store these containers, yard cranes at the storage area will serve the coming trucks and arrange containers. Each of these processes involves various decisions.

Regarding the sequencing for trucks, Kim et al. (2003) formulate a model for a single crane to minimise the integrated service costs related to time and distance. DP with a rolling horizon strategy (DPRHS) is designed for sequencing trucks, which arrive in a continuous fashion. Their proposed QL is compared with other traditional sequencing rules and outperforms them when the initial distribution of containers' arrival locations deviates from a uniform distribution. Fotuhi et al. (2013) study yard crane operations with RL. The authors model yard crane operators as MDP and apply QL to derive solutions. The learning objective is to locate the lowest reward, which is to minimise average truck waiting time. Their results show that the number of cranes and the truck arrival rate have a great impact on the performance. The study of Zeng et al. (2011) aims at minimising the time for loading or unloading a given set of containers. The authors propose a QL approach, which is divided into two problems: yard crane scheduling (Agent 1) and yard trailer (truck) scheduling (Agent 2). The first problem is to reduce the waiting time of yard trailers by optimising the sequence of cranes. Agent 2 deploys a QL algorithm and aims to reduce the waiting time of quay cranes. The solution method is further enhanced by integrating QL and simulation (RLSO). Their results show that RLSO performs the best in all nine methods examined in terms of relative deviation. However, with an increased number of quay cranes, the outperformance becomes mild. This conclusion is consistent with Kim et al. (2003)'s and Fotuhi et al. (2013)'s findings.

Chen et al. (2019a) study path planning and autonomous driving of cargo ships. Based on ship kinematics, the first-order Nomoto model is used to establish the initial Q -table. The first-order KT model (Zhang et al., 2012) determines the ship action and the state of the next step. In comparison among QL, Rapid-exploring Random Tree, and A* algorithms, the length of the path and the number of sharp turns are used as the performance measures. Their experiments suggest that QL has better performance than the others, especially when there are more obstacles in the scenario.

Another maritime application is spatial resource arrangement. Kim et al. (2020) minimise the rearrangements of the locations of ship blocks in the shipbuilding industry by a multi-agent A3C method. There are two decision processes successively involved: (1) arrangement of incoming blocks, determined by a locating agent (LA); (2) rearrangement of outgoing blocks, determined by a

Table 4
Summary of applications for maritime logistics..

Study	Method(s)	State	Action	Problem	Application
Kim et al. (2003)	Tabular QL	Waiting lines and trucks' information	Choice of the next served truck	Sequencing	Truck scheduling
Zeng et al. (2011)	Multi-agent, Tabular QL	Trailers' and cranes' information	Assignment of cranes to trailers	Sequencing	Truck and crane scheduling
Fotuhi et al. (2013)	Tabular QL	Cranes' and trucks' information	Choice of the next served truck	Sequencing	Crane scheduling
Irannezhad et al. (2017)	Multi-agent, DQN	Daily shipments	Decision strategies	Routing	Port community systems
Chen et al. (2019a)	Tabular QL	Coordinates of ships	Rudder angles	Routing	Cargo ship path planning
Li et al. (2019b)	Multi-agent, DQN	Status of the logistics network	Loading/unloading at terminals	Allocation	Ocean transport logistics networks
Irannezhad et al. (2020)	Multi-agent, ADP	Daily shipments	Decision strategies	Routing	Port community systems
Kim et al. (2020)	Multi-agent, A3C	Blocks in stockyard	Arrangement location of the next block, and movement of a transporter/loading of blocks to a transporter	Allocation	Ship stockyard arrangement
Farahani et al. (2021)	DQN	Containers' information and Trains' availability	Assignment	Allocation	Container assignment

transporting agent (TA). Each of them is an independent A3C DRL agent. At each LA's step, the action is to locate a block, while each episode gives an arrangement schedule for a specific period. As a result, before training LA, TA is trained to transport blocks to destinations with random initial blocks' locations. The well-trained TA is utilised in the training of LA.

Empty container repositioning is a similar application to spatial resource arrangement. Li et al. (2019b) develop a cooperative MARL method for a complex transport network in the ocean. They formulate the logistics network problem as a stochastic game. The three-level cooperative metrics of MARL are designed to describe the collaboration of agents, who are dependent on each other. The target is to minimise the accumulated resource shortage (as rewards) for all terminals.

Farahani et al. (2021) propose an online DQN algorithm for the matching of containers and trains. An agent considers the remaining capacities associated with the departure and arrival dates for each train, as well as the information about the container, to decide which train to deploy.

Table 4 summarises the maritime logistics applications we reviewed. These applications have characteristics of problem uncertainty and complexity under changing environments. Their problem natures suggest that RL is an appropriate approach to enabling the applications. These applications can be classified as sequencing or scheduling problems, path planning, and resource allocation. There are more exciting maritime applications that may be explored with an RL approach. For example, with a more dynamic environment affected by cyclones or ocean currents, the path planning and empty container repositioning problems become more uncertain and challenging. The various types of containers, freights, and cargoes may also impose further complexities in terminal operations. Another important aspect that may require more research efforts would be the sustainability aspect and environmental concerns in maritime logistics. For example, a potential application is that RL may be adopted to reduce pollutants and emissions generated by vessels during transport.

3.3.4. Urban logistics

Urban logistics is a broad topic, as different logistics companies and organisations may have their unique problem statements. Nevertheless, in terms of problem natures, many of the applications in urban logistics can be associated with a set of VRP variants, such as VRPPD, VRPTW, and CVRP. Other related applications include the shortest path problem, military logistics, and route preference evaluation. We remark that research studies *solely* focusing on the solution methods for solving VRP and its variants have been reviewed in Section 3.3.1 and are not discussed here.

The dynamic VRPPD is the most widely studied problem in the literature regarding the RL applications in urban logistics. It aims to deploy a fleet of vehicles to fulfil pickup and delivery demands at specific locations. In an RL setting, a typical approach is to model the transition process of a semi Markov chain whenever an order arrives. At each transition stage, there are three levels of decisions (from the highest to the lowest): order acceptance/rejection, job assignment, and path construction. Depending on the problem nature, some studies incorporate only one or two decision levels. For example, there are applications that require that all orders shall be accepted. In these applications, the decisions on order acceptance/rejection are irrelevant. Some other studies consider that couriers can pick up multiple items subject to their capacities, while others assume that orders have to be fulfilled once at a time. In addition to studies which recursively solve static PDPs, a few articles consider withholding orders or predicting future demands.

At the highest level of decisions, logistics companies or organisations can decide to accept or reject customer orders. In some problems, acceptance/rejection is the only decision suggested by RL; other decisions are made by optimisation algorithms. Kang

(2018) presents a DQN method to determine whether to accept pick-up and delivery orders with a single vehicle. In their proposed framework, the agent considers the location and weight of the parcel and information about the vehicle (location, schedule, battery level, and capacity) and then decides whether to accept the order or not. Their approach utilises only a single agent, and therefore no order assignment decisions are required. In a later paper, Kang et al. (2019) generalise the approach to the case of multiple vehicles. An intelligent order management system retrieves fleet and parcel information and decides whether to accept or reject an order for the entire fleet. Kavuk et al. (2021) solve an order dispatch problem which determines only acceptance/rejection decisions, while assignments are made by a greedy rule. Mo and Ohmori (2021) study a similar problem where drivers can reject orders and reassignment can be made in their simulation model. Manchella et al., (2020b, 2021, 2020a) apply a slightly different concept to another variant of the problem. They present a multi-hop joint passengers and goods delivery problem. In their setting, order acceptance/rejection decisions are made *after* order assignment. A central controller, enabled by DQN, assigns customer orders with a greedy policy; vehicles act as agents to decide acceptance of the assigned order; and routing is performed by a standard heuristic. Another concept is to determine acceptance/rejection and assignment jointly by modelling individual couriers as agents, who are trained to accept or reject an incoming order (Jahanshahi et al., 2021). Such an algorithm has a hard rule to refrain idling vehicles from rejecting the order. The work of Chen et al. (2020b) is also based on this concept, but incorporates fairness in deliveries between urban and rural areas into the objective function. In summary, acceptance decisions can be made before, jointly with, or after assignment.

At the intermediary level, once an order is confirmed, the next decision is to assign the order to a vehicle. One popular research topic is virtual bidding. In the related problems, each vehicle is modelled as an agent who determines a virtual bidding price, taking into account information about the requests, price, and inputs from other agents. The objective of bidding is to maximise the overall order fulfilment rate and save the travel costs. Chen et al. (2019b) study this problem using Proximal Policy Optimisation (PPO) algorithm. They then extend the work and propose a QMIX ‘DeepFreight’ algorithm which allows multi-vehicle transfers (Chen et al., 2021b). Li et al. (2021) propose a similar bid-to-assign approach with a neighbourhood attention mechanism, coupled with GNN, to solve an industrial scale problem. In addition to the RL model, the authors train another NN to predict the future spatial and temporal demand at each factory. Another stream of research instead jointly optimises routes and assignments. Several research studies integrate heuristics and RL methods to tackle the problems. Ahamed et al. (2020, 2021) train a DQN agent to choose heuristics, including exchanging and inserting nodes within and between carriers, according to the delivery instruction. In addition to the embedding of heuristics, they design a Tabu list and a priority list to prevent agents from repeatedly producing the same sequence and to encourage exploration. Ma et al. (2021) study a problem which breaks a dynamic PDP into static sub-problems. They extend a heuristic RL method to a HRL framework. All upcoming orders are stored inside a buffer, and are released at once when the upper agent instructs so. The lower agent selects heuristic operators to solve static VRPPD. Different from Ahamed et al. (2021), they apply a Graph Isomorphism Network, which is trained with a PG algorithm. There are research studies which do not adopt main stream methods to solve the assignment problem. Instead of determining assignment with an RL agent, James et al. (2019) propose an algorithm to derive routes. An information centre assigns different delivery tasks recursively to each vehicle until an optimal combination is identified. In the routing decisions, each vehicle is modelled with a STRUCT2VEC GNN for path construction. Zou et al. (2021) propose a Double DQN model which considers the locations of customers, couriers and estimated arrival times. The model directly assigns each task to individual agents. Besides making specific assignments of orders to resources, RL is also capable of determining assignment decisions at a tactical level. For example, Hu et al. (2021) cluster a city into discrete regions and propose an Actor-critic algorithm to deploy couriers. Reward is defined as the ratio of the number of orders to the number of couriers in the region. In summary, for assignment decisions, strategies such as bid-to-assign, direct assignment, and recursive trial assignment have been explored. Hybrid RL and heuristics are popular methods to deliver solutions.

At the lowest level of decisions, after assigning orders to vehicles, the routes of the vehicles have to be constructed. On an urban motorway, traffic conditions usually exhibit diurnal patterns and uncertain outcomes. Most of the existing studies rely on the least expected time or risk-based methods to handle stochasticity (Nie and Wu, 2009). Cao et al. (2020) tailor a DQN algorithm for this problem, where agents can choose their directions according to the Q -value at each junction. It is worth noting that this algorithm utilises historical traffic data. Ding et al. (2021) present a QL algorithm for multi-hop public-transport-based crowdsourced delivery taking into account public transport systems’ conditions. Their DQN model learns the information about the parcels and passengers to determine dispatching decisions. This distributed approach is different from previous studies of multi-hop delivery which only consider acceptance or assignment.

In contrast to dynamic VRPPD, which requires fast solutions, VRPTW is designed for planning purposes and can tolerate longer computational times. One popular approach is to construct an RL agent to determine the routes and assignments directly. Zhang et al. (2020a) study a VRPSTW and adopt a transformer architecture as the RL agent, trained with a PG method. The transformer encodes the customer information with embeddings, and then decodes with multiple decoders representing individual vehicles. Qiu et al. (2021) study a slightly different problem variant, called ‘deterministic home delivery and installation’, with PG-trained transformer architecture. Vehicles and crews travel respectively to customer locations. Different from other studies, their approach does not require a separate assignment stage as a minimal usage of vehicles is encouraged. Another stream of research entails combinations of RL and heuristics, in line with the observations we had in the literature regarding the assignment decisions in VRPPD. Silva et al. (2019) propose an RL-based metaheuristic for solving VRPTW and parallel machine scheduling problems. Their algorithm consists of several independent local search algorithms as agents, who communicate with each other. Each agent performs an adaptive local search through several operators such as swap, exchange, and insertion. Each of the agents constantly updates the Q -table until convergence. A similar study by Chen et al. (2020a) solves a periodic VRPTW. They first use an urgency-level based insertion tactic to construct initial solutions and then develop an RL-based local search approach to refining the solution quality.

The local search adaptively updates the probabilities of operators at each iteration in the form of online RL. Similarly, a heuristic RL tactic is proposed for CVRP by [Martin et al. \(2016\)](#). A launcher initiates a user-defined number of metaheuristics and tunes the hyperparameters. The metaheuristics communicate with each other at a predefined frequency by sharing their permutation edge information. Frequently appeared edges are stored in a short memory pool for future execution. We remark that the studies found in this stream of research all utilise online RL optimisation methods, which do not require prior training. On the other hand, in VRPPD, studies adopt offline RL methods.

In the reviewed applications for urban logistics, losses or damages of items are rarely considered. An interesting study presented by [Stimpson and Ganesan \(2015\)](#) considers deliveries under potential attacks in a military setting. In their study, there are route clearance vehicles that can detect the presence of improvised explosive weapons. A value-based learning agent takes into account the customer inventory level, readiness of clearance vehicles, occurrences of attacks on each route from historical data, and then decides the routes, convoy fleet size, and the deployment of clearance vehicles. The objective is to minimise the risk of vehicles being attacked and the operational costs, while the reward related to delivery is maximised.

Another study that presents unique problem characteristics under urban logistics considers route preferences of delivery persons. [Liu et al. \(2020\)](#) aim to address the issue that the shortest path approach may not be effective in route planning for bike/motorcycle deliverymen due to incomplete road network information. They suggest learning from historical data and propose a Dijkstra's algorithm-based IRL approach to determine routes. This algorithm considers the state (discretised location blocks) transition policy based on real data and the paths suggested by Dijkstra's algorithm.

In conclusion, most research studies present applications for urban logistics at different decision levels. The majority of these studies develop RL approaches for decisions at only a single level, and applies heuristics or exact algorithms for decisions at other levels. Integrated approaches of RL and heuristics are prevalent in this research area and have demonstrated tradeoffs between solution effectiveness and computational efficiency. [Table 5](#) summarises the RL applications for urban logistics.

While urban logistics is one of the main areas that RL has been applied, there is still significant room for future research. For example, given the different levels of decisions under an urban logistics setting, a sensible way would be the adoption of hierarchical RL approaches that decompose the overall problem into smaller subtasks and train multiple layers to determine decisions at different levels ([Nachum et al., 2018](#)). Further research would be essential to examine the effectiveness of these approaches. In urban logistics applications, the utility of couriers and customers is an important aspect. IRL approaches, which are rarely adopted in urban logistics applications, would be a promising research direction due to their nature of learning from historical data regarding the courier and customer preferences.

We remark that an interesting, yet inadequately studied, topic is military logistics. The problem nature of military logistics shares similarities with disaster relief, which will be discussed in [Section 3.3.9](#), as the problems in both areas consider potential losses of goods and items due to sabotage. In our review, only one research study is found to be related to military logistics. Different objectives, such as maximisation of items delivered and minimisation of losses, overstocks, and delivery costs, would be potential considerations in military applications.

3.3.5. Facility logistics

Facility logistics is concerned with the movements and handling of parts, products, materials, parcels, and other items within a facility (e.g., factory, warehouse, and distribution centre). The efficiency of a logistics system in a facility greatly determines the responsiveness to customer demands. Therefore, optimisation of facility logistics not only leads to significant operational cost savings, but also enhances customer experience and induces higher future demands. Traditional optimisation approaches, such as mathematical modelling and heuristics, have been developed to suggest good facility logistics decisions. Since the environment in a facility typically is highly dynamic, fast and high-quality solutions are desired for real-time implementation. The recent advance of RL has opened up a new research direction to achieve the goal.

Among the studies on facility logistics, most of them fall within the applications of RL for delivery of goods or other items in a warehouse. These applications have become more popular nowadays due to the prevalence of E-commerce. The essence of the optimisation of delivery operations in a warehouse not only results from the greater volumes of products required to handle, but also addresses the increased number of handling operations per order due to the product heterogeneity within an order and across different orders. In this type of problems, most of the studies consider the deployment of unmanned vehicles (UVs) for delivery purposes. From our review of these studies, we observe that, similar to VRPPD, there are two levels of decisions, where the higher level is order assignment and the lower level is related to picker routing optimisation.

At the higher level, customer orders are assigned to different UVs. [Malus et al. \(2020\)](#) use a Twin Delayed DDPG algorithm to solve an UV bid-to-assign problem. Multiple agents (UVs) sharing the same objective and set of parameters bid according to the locations and deadlines of orders. In addition to bidding, heuristic actions have also been proposed to assign orders. [Hu et al. \(2020\)](#) propose a DQN model, where multi-layer perceptrons act as an agent to choose heuristic operators to assign orders. Their algorithm can also be applied to materials handling in a factory ([Govindaiah and Petty, 2019](#)). If the problem considers workers only, then only assignment decisions are determined. For example, [Estanjini et al. \(2011\)](#) propose a parametric model, trained with a least square TD learning method, for a manned forklift operations problem. The model takes the inventory information and infrared sensing information from the liftforks, including the collision, battery level and current location, and assigns the liftforks to specific tasks.

At the lower level, once the order assignment is determined, UVs are deployed to various locations within the facility to pick up items of the orders. If the destination is known to the vehicle, a shortest path algorithm may be used to guide the vehicle. Different from a simple routing problem, a routing algorithm in a warehouse should avoid vehicle collisions, while simultaneously

Table 5
Summary of applications for urban logistics.

Study	Approximator(s)	Method(s)	Problem	Problem solved by RL
Stimpson and Ganesan (2015)	Single DNN	Value-based learning	Military logistics	Routing + Fleet deployment
Martin et al. (2016)	Metaheuristic agents	Improvement heuristic	CVRP & Permutation flow-shop scheduling problem	Routing + Assignment
Bono et al. (2018)	No agent	No learning	VRPPD	N/A
Kang (2018)	Single DNN	QL	SVRPPD	Order acceptance
Chen et al. (2019b)	Multi DNNs	PPO	MVRP	Assignment
James et al. (2019)	GNN	PG	VRPPD	Routing
Kang et al. (2019)	Single DNN	QL	MVRPPD	Order acceptance + Routing
Silva et al. (2019)	Q-table	QL	VRPTW	Heuristics selection
Ahamed et al. (2020)	Single DNN + Heuristics	QL	MVRPPD	Heuristics selection
Cao et al. (2020)	Single DNN	QL	Stochastic shortest path	Routing
Chen et al. (2020a)	Metaheuristics	Improvement heuristic	Periodic VRPTW	Heuristics selection
Chen et al. (2020b)	Single DNN	QL	MVRP	Order acceptance
Liu et al. (2020)	Single DNN	IRL	Inference of user preference	N/A
Manchella et al. (2020a)	Single DNN	QL	Dispatch of taxis (for goods & passengers)	Locations to dispatch
Manchella et al. (2020b)	Multi DNNs	QL	Joint passengers and goods delivery	Pricing + Assignment
Zhang et al. (2020a)	Transformer	PG	VRPSTW	Routing + Assignment
Ahamed et al. (2021)	Single DNN + Heuristics	QL	MVRPPD	Heuristics selection
Ding et al. (2021)	Multi DNNs	QL	Dynamic PDP	Routing + Assignment
Hu et al. (2021)	Single DNN	AC	Dispatch of couriers	Locations to dispatch
Jahanshahi et al. (2021)	Multi DNNs	QL	MVRPPD	Order acceptance + Assignment
Kavuk et al. (2021)	Multi DNNs	QL	MVRPPD	Order acceptance
Li et al. (2021)	Spatial-temporal aided double deep graph network	PG	Dynamic PDP	Assignment
Ma et al. (2021)	DNN	HRL (QL + PG)	Dynamic PDP	Order withholding + Heuristic selection
Manchella et al. (2021)	Multi DNNs	Double QL	Joint passengers and goods delivery	Order acceptance
Mo and Ohmori (2021)	DNN	QL	Crowdsourced dynamic PDP	Order holding
Chen et al. (2021b)	DNN + GRNN + MILP	QMIX	Multi-vehicle transfer PDP	Assignment
Qiu et al. (2021)	Transformer	PG	Home delivery and installation	Routing of crew and vehicles
Zou et al. (2021)	Single DNN	Double DQN	Dynamic PDP	Assignment

coordinating a large number of vehicles. Kamoshida and Kazama (2017) and Li et al. (2019a) respectively propose DQN models specifically tailored for this problem. Their algorithms take the pixelated map as an image and input it into the CNN, and then route each vehicle at discretised time intervals. The former uses a soft constraint for collision avoidance, while the later uses a hard rule. A later study conducted by Zhang et al. (2020b), who adopt a similar idea but a multi-step ahead tree search strategy, is proven to have even fewer collisions and less congestion. In addition to the prevailing studies of model-free methods, Andersen et al. (2019) apply a model-based algorithm, Dreaming Variational Autoencoder, to solve a multi-agent path finding problem. This algorithm has another module to learn the transition probabilities after taking certain actions, and is demonstrated to perform better than DQN or PPO. Hwang and Jang (2020) cast doubt on the scalability of the multi-vehicle agent approach, and instead use nodes as the agents. They propose a $Q(\lambda)$ -learning approach to solving a dynamic overhead hoist routing problem in a factory setting. Their algorithm treats the nodes as RL agents, which consider the destinations of the incoming vehicles, and assigns the next node for vehicles. Ahn and Park (2021) consider the operations of idling vehicles. Each zone is modelled as an agent, who decides to send off hoists inside the zone to neighbouring zones. GNN is used for extracting the embedding of each zone. Last but not the least, a more realistic QL-based shortest path algorithm is proposed by Sun and Li (2020) to solve UV routing problem. An agent takes camera images, laser information, and locations as states and then decides the speed and angular velocity of an individual vehicle.

Table 6
Summary of applications for facility logistics.

Study	Environment agent(s)	State	Method(s)	Problem
Estanjini et al. (2011)	Central controller	Inventory levels and sensors' information	Least squares TD + AC	Forklift dispatching
Kamoshida and Kazama (2017)	Individual vehicles	Images and sensors' signals	DQN	Individual routing
Andersen et al. (2019)	Central controller	Maps	PG	Group routing
Govindaiah and Petty (2019)	Heuristic selector	Current solutions	Tabular SARSA	Material transfer routing
Li et al. (2019a)	Central controller	Maps	DQN	Group routing
Cals et al. (2020)	Central controller	Orders' information	DQN	Order batching and scheduling
Hu et al. (2020)	Heuristic selector	Task workloads, deadlines, average distances, and vehicles' working statuses	DQN	Job assignment
Hwang and Jang (2020)	Junctions	Origin–Destination and current vehicle locations	$Q(\lambda)$ -learning	Group routing
Malus et al. (2020)	Vehicles	Orders' information	Twin Delayed DDPG	Job assignment
Zhang et al. (2020a)	Vehicles	Maps	Value-based approach	Group routing
Ahn and Park (2021)	Zones	Number of vehicles	AC + GNN	Rebalancing vehicles
Cals et al. (2021)	Central controller	Orders' information	PG	Order priority
Ni et al. (2021)	Human pickers/robots	Gantt chart	PG	Hybrid flow shop scheduling
Rim���� et al. (2021)	Central controller	Shelves and robots' information	Double differential DQN	Shelf allocation
Yu et al. (2021)	Vehicles	Images and Origin–Destination	DQN	Individual routing

Table 7
Summary of applications for logistics demand prediction.

Study	State	Action	Problem
Hassan et al. (2020)	Spatial–temporal pattern	Predictor selection	Demand prediction

Regarding order assignment with both human pickers and UVs, the reader may refer to the study of Cals et al. (2020), which provides a comprehensive approach to an order assignment problem. They craft a DQN model for solving an order batching and sequencing problem. The RL agent captures the information about remaining orders, availability of resources, tardiness, and processed orders into account to determine whether a part-to-picker mode or a picker-to-part mode to adopt. More recently, the same group of researchers further developed their model using a PPO algorithm to directly control which orders to pick according to order classification (Cals et al., 2021). A similar problem is investigated by Ni et al. (2021). They propose an RL and heuristic combined algorithm for a static hybrid flow shop scheduling problem. They turned the Gantt chart of jobs into multiple directed graphs, then used a GCN neural network, trained with PG, to determine the heuristic operators.

Recent research has been conducted to deploy robots to move shelves at warehouses. Rim     et al. (2021) propose a Double Differential DQN to decide the putaway locations of the shelves based on minimal expected cycle time.

Table 6 summarises the applications of RL for facility logistics. The articles in this subsection typically consider problems at a single decision level. At the other level, either assumptions are made or heuristics are deployed. Through our literature review, there is no study which addresses problems with RL at these two levels simultaneously. HRL is presumably an unexplored, yet appropriate, choice for facility logistics. However, the algorithmic efficiency depends heavily on the problem nature, and one algorithm may not be applicable to another problem.

3.3.6. Logistics demand prediction

Demand prediction is an important topic in logistics systems in practice. While accurate forecasts of demands are beneficial to effective decisions to match the supplies with the demands, surprisingly, our literature search could only identify one research study on this topic. Hassan et al. (2020) propose an RL approach to freight demand forecasting. In the first step, the market demands are clustered according to the spatio-temporal patterns and commodity types. In the second step, freight demand, measured in terms of the number of containers moved, is forecasted with several existing models in different time horizons. The RL agent learns from the historical data and assigns weights to the model predictions.

From our review, this research area is not well studied. We believe that the inadequate research on this topic is due to the fact that RL requires an underlying environment for agents to interact with, but demand prediction is heavily driven by historical data where an RL environment is difficult to be constructed. We remark that the only study we have identified deploys the RL agent just to learn the weights of different prediction models, instead of learning the demands directly (see Table 7).

3.3.7. Pricing, bidding, and negotiation in logistics

In contrast to the studies on logistics applications with full control over a centralised fleet, a number of papers study optimal bidding in the spot market of the freight transport industry. Instead of having fixed long-term contracts with logistics providers, all the order assignments are dynamic and price-dependent. In the spot market, shippers have the delivery requirement and pay carriers

to perform the delivery services; carriers bid to receive orders from shippers and aim to gain profits; administrators are obliged to ensure the pollutants generated during transport staying within a certain level and avoid congestion as possible; urban consolidation centres (UCCs) gather all the shipping requirements and deliver the orders in bulk; and brokers gain profits from information asymmetry in the spot market (Cleophas et al., 2019). For a comprehensive review of freight logistics service procurement, we refer the reader to Lafkihi et al. (2019).

The first stream of research articles model shippers and carriers as RL agents. Taniguchi et al. (2007) propose a model where carrier agents apply DQN to bid with discrete price intervals to maximise profit. Carrier agents may use exact VRPTW-Dynamic or VRPTW-Forecasted models to obtain the routes. Shipper agents minimise the sum of delay penalties and service charges, where the delay penalties are learned from carriers' performance. The administrator agent learns to determine the emission charge on each road. Guo et al. (2021a) study a simplified version of this problem. Carriers can only decide to give low, medium, or high prices. They offer a comparison of DQN, Tabular QL, and DQN with experience replay. In van Heeswijk (2021), carriers and shippers are given the expected prices of the counterparty. Both parties can submit their requests to a broker, who matches the jobs and earns difference between two prices. Carriers and shippers are modelled with a PG approach and decide whether to accept the current arrangement.

The second stream of research focuses on models administrators as RL agents. Carriers' pricing depends only on the delivery costs; shippers select the carrier according to a second price auction principle. Teo et al. (2012a) study a problem where administrators learn the road toll charges to minimise pollution. Teo et al. (2012b) study the use of cordon-based pricing, where vehicles are charged as they enter urban areas. Teo et al. (2014) extend the research considering load factor pricing and distance pricing imposed by the administrator, as well as a few other performance indicators. The administrator learns the charging rate, while the carrier can learn the bidding price. The same group of authors study the offer of subsidy and parking fee waivers to UCCs operators with a QL approach to reduce emission (Wangapisit et al., 2014).

The third research stream focuses on the benefits of UCCs. In Firdausiyah et al. (2019), a freight carrier is modelled as an RL agent (with an ADP approach). The agent can choose to deliver the freight directly on their own or dedicate the service to the UCCs with a certain payment. UCCs is another RL agent that decides the service charge in order to maximise the profit. The update rule of the parameters of the agents is based on ADP, where the states are the customers' demands and time windows. In their later studies (Firdausiyah et al. 2017, 2018, 2020), similar approaches are developed study simplified versions of this problem and ADP is shown to be a superior solution in all scenarios.

There are studies which consider other parties or entities in a pricing, bidding, or negotiation setting. Wong and Ho (2010) propose a QL-based approach for a problem in an open railway access market, where an infrastructure provider receives negotiation requests from a number of train service providers and decides time tables. Both the train service providers and the infrastructure provider aim to increase their profits during the negotiation process; the infrastructure provider can decide to increase or decrease the charging levels; and the train service providers can change the service start times. Wang et al. (2018) use a broker to decide the price and assignment of tasks, and extract profit from a portion of the agreed price. Shippers and carriers can only decide if they accept or not the price, without a need to propose any service charge. The authors adopt knowledge gradient, a Bayesian optimisation method, with a logistics function setting to address the problem of incomplete learning, where a transition may lead to the same cumulative reward. In van Heeswijk (2020), containers are modelled as intelligent bidding agents (with a PG method), whose objective is to meet the delivery deadline at a minimal cost. Carriers decide which items to deliver according to their bidding prices. The state information amongst containers can only be known if an agent shares its own. The author concludes that no information sharing yields similar performance with full information sharing while partial information sharing performs the worst.

Table 8 summarises the applications for pricing, bidding, and negotiation in logistics. In summary, the research studies reviewed under pricing, bidding, and logistics address slightly different variants of the spot market problems in the freight transport industry. The studies consolidate the existing pricing methods in the industry with RL approaches and provide transferable frameworks for the parties in the freight market. We have identified several future research opportunities. First, as previously discussed, discrete values of price may not provide a comprehensive exploration of the pricing decisions for the best objective. Second, the costs of maintaining and operating a fleet of delivery resources in a market with UCCs are rarely considered in the literature but have great impacts on decisions in practice. Third, some assumptions, such as constant vehicle speed, deterministic travel times, and uniform arrivals of customer orders, may be relaxed for more practical applications. Finally, the use of real-world data would further demonstrate the practicality of the research and provide non-trivial insights into addressing realistic problems.

3.3.8. Resource allocation

In logistics systems, resource allocation requires the distribution of resources to different demand points to enable the required services or processes. Typically, demand forecasting is required for allocation decisions. Traditionally, there can be a two-phase approach where demand prediction is conducted first to supply input parameters to an optimisation model for decision-making. The development of RL offers an alternative and effective way to offering decisions, where demand prediction and decision-making are jointly carried out.

Bike repositioning is a successful example of RL for resource allocation. In the literature, most of the bike repositioning problems are solved by exact algorithms or heuristics. Recently, there have been promising applications of RL for bike repositioning. Li et al. (2018) present an offline multi-agent DQN model for docked shared bike repositioning using trikes. At the first stage to cluster regions with the aim of maintaining inner-balance (having similar bike rent and return demands within a cluster) and inter-independence (having only small bikes between two clusters). At the second stage, a DQN trike agent considers the current numbers of bikes at the stations, predicted demands, time, weather, and information about other trikes and itself to decide the quantity of

Table 8

Summary of applications for pricing, bidding, and negotiation in logistics.

Study	Environment agent(s)	Method(s)	Continuity of prices	Insights
Taniguchi et al. (2007)	Carriers, shippers, and administrators	DQN	Discrete	VRPTW-D is better than VRPTW-F
Wong and Ho (2010)	Infrastructure and train service providers	DQN	Discrete	The proposed algorithm improves profit
Teo et al. (2012a)	Administrator	DQN	Not mentioned	Intelligent road tolls can reduce pollution
Teo et al. (2012b)	Administrator and carriers	DQN	Ratio	Joint distance and load factor charging is better than pure distance charging
Wangapisit et al. (2014)	UCC operators, carriers, and administrator	DQN	Discrete	Subsidising UCCs can reduce emission
Teo et al. (2014)	Administrator	DQN	Ratio	Cordon-based charging escalates the delivery costs
Firdausiyah et al. (2017)	Carriers	ADP + DQN	Discrete parking charges + exact operational costs	ADP is better than DQN
Firdausiyah et al. (2018)	Carriers	ADP + DQN	Discrete parking charges + exact operational costs	ADP is better than DQN
Wang et al. (2018)	Broker and Carriers	Bayesian methods	Continuous	Knowledge gradient is better than others
Firdausiyah et al. (2019)	Carriers and UCC operators	ADP + DQN	UCCs: up, down, or same; Carriers: discrete parking charges + exact operational costs	ADP is better than DQN
van Heeswijk et al. (2019)	UCCs	ADP	Continuous	ADP is better than heuristics
Firdausiyah et al. (2020)	Carriers and UCC operators	ADP + DQN	UCCs: up, down, or same; Carriers: discrete parking charges + exact operational costs	ADP is better than DQN
van Heeswijk (2020)	Smart containers	PG	Continuous, Gaussian distributed	Smart containers are efficient
Guo et al. (2021a)	Carriers	DQN + Tabular QL	Discrete	Learning is better than no learning
van Heeswijk (2021)	Carriers and shippers	PG	Continuous	High uncertainty leads to systems inefficiency

bikes to unload or load at a certain location (one location at a time). Pan et al. (2019) solve a bike repositioning problem by offering monetary incentives to users to balance the bike distribution, using an HRL algorithm built upon DDPG. Each sub- $Q(\lambda)$ agent, representing a small region, learns the sub- $Q(\lambda)$ values from the historical demands and demands in neighbouring regions, and then determines the price. The Q -agent at the higher level aims to maximise the users' demand fulfilment subject to an incentive budget.

Besides bike repositioning, Rabe and Dross (2015) study the distribution of resources amongst warehouses, based on RL and an external simulator. The simulator retrieves inventory data from warehouse management systems. The RL model minimises the overall costs of delivery in the whole network.

Table 9 summarises the applications for resource allocation. Resource allocation is an area that lacks sufficient research, in terms of both methodology and application. First, it is worth enhancing the simulation environment to make it more realistic. For example, general utility functions may be explored to enhance the practicality of the model. Second, it would be interesting to explore an integrated problem of pricing and bike rebalancing. Third, from a methodological perspective, multi-agent cooperative RL is another plausible learning method to explore. Finally, the RL techniques developed for bike repositioning, may be transferrable to solve other types of problem in shared mobility systems, such as relocation problems in carsharing systems (Martin et al., 2021; Martin and Minner, 2021).

3.3.9. Disaster relief

Rescuing lives after disasters involves various requirements, such as satisfying the urgent needs of affected areas, ensuring short response time to complete the missions, and offering effective and fair humanitarian services. The challenges of disaster relief

Table 9

Summary of applications for resource allocation..

Study	Method	State	Action	Problem
Rabe and Dross (2015)	Tabular QL	Not clear	Not clear	Resource distribution amongst warehouses
Li et al. (2018)	DQN	Number of bikes, demands, trike positions, and loads	Load or offload at a location	Bike redistribution with trikes
Pan et al. (2019)	HRL QL	Historical demands and demands in neighbouring grids	Price	Pricing for every region

Table 10

Summary of applications for disaster relief.

Study	Method	Problem setting	Problem
Nadi and Edrissi (2016)	Tabular QL	Road failures, dynamic demands, one vehicle, softmax action selection	Delivery of items to affected regions
Nguyen et al. (2018)	Heuristic MARL	Locations of rescuers and victims	Resource allocation and matching
Yu et al. (2021)	DQN	Consideration of costs but not routing	Coordination of items for rescuing population in disaster areas

include collection of real-time disaster-related data, uncertainties arising from disaster damages, complexities of operations, and construction of the right objectives to address the needs. The reader is referred to Galindo and Batta (2013) for a comprehensive review of disaster relief research conducted with general optimisation techniques. Through our literature search, we find that the applications of RL for disaster relief share similarities of methodologies adopted for resource allocation, presented in Section 3.3.8. RL has the advantage of offering fast solutions under uncertainty and therefore is well suited to this class of problems. The studies on disaster relief can be summarised as assignment and routing problems, where problem-specific constraints and objectives are imposed.

Nadi and Edrissi (2016) provide one of the earliest studies on disaster relief. The authors propose an online tabular QL method to construct vehicle routes for distribution, with a small probability of not reaching the destination due to road damage. Nguyen et al. (2018) design a ResQ algorithm for the matching of rescuers and victims. A QL algorithm considers the geolocations of victims and carries out matching by choosing heuristics as actions. Yu et al. (2021) provide another QL framework for resource allocation. A local response centre shall assign the available items to regions with specific demand quantities.

The applications on disaster relief reviewed in this subsection include vehicle routing and the matching of resources to demand points. Methodologies are mainly QL-based. Agility (short relief time) and successful rescue rate are typical key factors to develop a disaster relief plan.

Table 10 summarises the applications for disaster relief. There are some future directions of disaster relief research. First, multi-modal transport (drones, trucks, and rails) has great potential to provide more effective and agile humanitarian services. For example, it would be exciting to study the adoption of drone surveying aided vehicle delivery and drone delivery. Second, in an environment affected by disaster, high levels of uncertainties arise, and the transport infrastructure is typically damaged. Only partial information is known to the decision-maker. In such a situation, RL may have the advantage to learn from the disaster information to determine the best decisions. Third, joint decision-making in different problems would be necessary for an integrated planning, such as synchronisation and coordination of vehicles and personnel (Dubey et al., 2019), and integrated resource allocation and routing. From the methodological perspective, RL architectures developed for taxi re-positioning and bike re-positioning, such as HRL and GNN, are transferable to disaster relief operations. The adoption of these architectures in disaster relief would be a potential new research direction.

3.3.10. Electric vehicle deployment

In recent years, electric vehicles (EVs) have been prevalent due to certain advantages or features. For example, EVs are environmentally friendly, and electric motors can be deployed to satisfy needs promptly. Therefore, EVs are expected to replace conventional vehicles in both transport and logistics networks. However, one major drawback of deploying EVs is that charging an EV is often not as convenient as refuelling conventional vehicles (He et al., 2016), and so it brings potential inconvenience to logistics providers. To this end, researchers have been studying different solutions for EV charging problems from different perspectives.

Some research focuses on the optimisation of the EV charging operations. In Asadi and Nurre Pinkley (2021), EV charging operations are modelled as an MDP. At each time step, an agent decides the number of batteries to charge or discharge, as well as the number of batteries to be replaced in a certain time interval. Their proposed model is solved by value iteration with a linear function approximation, as well as a heuristic benchmark policy. Another important problem related to EV charging is pricing. Fang et al. (2021) study a dynamic pricing problem for a fast-charging service. By considering the price elasticity (i.e., how customers react to the change of price), their problem is formulated as an MDP. The authors demonstrate how this problem can be solved using various RL algorithms, such as value iteration, QL, and AC algorithms.

Table 11
Summary of applications for electric vehicle deployment.

Study	Method(s)	State	Action	Problem
Chiş et al. (2017)	Fitted Q -iteration algorithm	Day, battery levels, and electricity prices and fluctuations	Amount of energy to charge the EVs	Scheduling
Wang et al. (2020)	Dyna- Q RL	Time, battery levels, energy prices, and possibilities of energy shortage	Discharge, idle, or charge the EV	Scheduling
Asadi and Nurre Pinkley (2021)	Approximate value iteration	Total number of full batteries and the average capacity of all batteries	Number of batteries to charge, discharge, or replace	Machine replacement
Fang et al. (2021)	DP, DQN, and AC	Price, consumed volume, available chargers, chargers' utilisation rate, traffic conditions, and alternative choice for customers	Pricing strategies	Pricing

Table 12
Summary of applications for railway systems.

Study	Method	State	Action	Problem
Khadilkar (2018)	Tabular QL	Adjacent trains	Halt or run	Train operations

From the perspective of an EV owner/driver, Chiş et al. (2017) consider the scheduling problem of charging or discharging EVs. The decision-maker determines the amount of energy to charge the EV, and the objective is to reduce the long-term electricity cost. A Bayesian NN is proposed to simulate the future prices, and a fitted Q -iteration algorithm is used to solve this problem. Wang et al. (2020) study a similar problem as in Chiş et al. (2017). The Q -function of the proposed MDP is approximated by a NN, and the problem is solved by Dyna- Q RL.

Table 11 summarises the applications for EV deployment. In summary, battery replacement operations and pricing of EV charging services are two main directions. The integration of these problems, e.g., dynamic pricing to influence EV charging demands for higher flexibility in producing battery swapping and charging schedules, would be an interesting problem to study. However, computational challenges arising from the higher complexity of the problem and the way to define the states and actions in an RL framework would require further research efforts.

3.3.11. Railway systems

Besides trucks, vessels, and aeroplanes, rail is another important means of delivering goods. In many countries, the delivery cost per tonne using rail transport is significantly lower than trucks. This subsection provides a review of the existing literature about RL in the delivery of goods by rail.

Due to the unique nature of rail transport, only two related studies are identified. Khadilkar (2018) propose a Q -tabular learning-based algorithm for train scheduling. A Q -table mimics an individual train, which captures information about the resources relative to the direction of the train's motion as the states and decides to halt or run the train when it arrives at a station or a new track. The other study is more related to the assignment instead of scheduling (Farahani et al., 2021). For the review of this study, please refer to Section 3.3.3.

We have identified further studies that do not exactly lie within the scope of this review but provide inspiring ideas for future research in logistics applications. One of the streams in railway research is energy-efficient train operations. For example, Yin et al. (2016) and Liu et al. (2018) respectively propose algorithms for optimal train scheduling with energy minimisation through parametric RL approaches. Recently, there are studies on the balance between energy costs and passenger demand satisfaction through multi-layer perceptrons (Ying et al., 2020, 2021).

In conclusion, since most of the research studies on railway applications are related to passenger transport, signal control, or energy optimisation, they are not reviewed in our work. There is a clear gap that researchers may explore the applications of RL for logistics systems operated by railways (see Table 12).

As a conclusion of Section 3.3, we summarise the typical characteristics in all logistics applications reviewed, as presented in Table 13.

3.4. Supply chain management

In this section, we review applications of RL for SCM. According to the problem nature, articles are classified into three categories. The subsection of *inventory control* is concerned with inventory decisions which interest and are optimised by only a single party. The subsection of *pricing* is relevant to the applications where the prices of products or services are to be determined. The subsection of *supply chain coordination* is concerned with the optimisation of the entire SC. We consider Vendor Managed Inventory (VMI) an application of SC coordination, as it offers mutual benefits to both retailers and upstream distributors.

Table 13
Summary of typical characteristics in logistics applications.

Category	State	Action	Reward
Air	Maps and demands	Assignment and routing	Distances and energy
Disaster relief	Maps, demands, and resources	Assignment	Shortages and costs
EV deployment	Battery levels, charging times, and energy prices	Pricing and scheduling of EV charging	Battery levels and costs
Facility logistics	Orders' information	Assignment and routing	Distances, penalties, and fairness
Logistics demands	Demands	Model weight estimation	Prediction accuracy
Maritime	Maps, congestion levels, and demands	Assignment, routing, and sequencing	Waiting times and distances
Pricing	Orders' information	Acceptance and pricing	Profit
Railway	Trains' information	Halt or run	Train delay, robustness, and fairness
Resource allocation	Maps, demands, and resources	Assignment	Shortage and costs
TSP and VRP	Maps and locations	Next node to visit	Distances and penalties
Urban logistics	Orders' information	Assignment and routing	Distances, penalties, and fairness

3.4.1. Inventory control

In our literature search, there are a number of articles on inventory control, or joint inventory control and pricing decisions. Among these papers, the problem natures and methods adopted are diverse. The simplest variant of inventory control only involves two tiers of parties. Each retailer is modelled as a learning agent, who decides order quantities to be placed to the supplier. Some scholars extend this setting to a multi-tier setting where multiple parties are involved. For example, in a four-tier SC, there can be manufacturers, distributors, warehouses, and retailers. The variants of the problems, models, and solution methods will be discussed in detail.

Many of the studies consider retailers' decisions on inventory control. Ravulapati et al. (2004) offer an early research study which applies RL for a SC inventory planning problem. They consider the problem as a multi-player non-zero-sum stochastic game and model it as a competitive MDP. In their RL framework, there are finite states, discretised actions, and a bounded reward. The Nash equilibrium (Nash, 1951) is used to evaluate the rewards of the players. Jiang and Sheng (2009) also consider a two-echelon SC, where multiple customers and retailers are involved in the two tiers, apply a case-based RL method to solve the problem. Each agent learns its current inventory level and average demand, and decides to adjust its price or reorder from suppliers. Similar to QL, the case-based learning discretises the state and action space into fine grids, and then the closest state shall be selected and updated during a training episode. The only difference is that the case-based reasoning method learns the reward before the lead time, whereas a QL model learns afterwards.

Kim et al. (2010) propose a MARL model for a multi-stage SC problem. Each retailer agent in the model applies a linear regression model for downstream demand prediction, and then incorporates the predicted value to determine a safety lead time. Once the predicted inventory level falls below zero in this safety lead time, a fixed size order is placed. Dogan and Güner (2015) study monopoly and duopoly retailers' markets with RL models. Retailers are modelled as RL agents who order from upstream manufacturers, and then determine the selling price. Customers are modelled with utility functions. It is found that a monopoly retailer sells the products with the maximum customer acceptable price. For a duopoly market, an RL model outperforms a traditional adaptive pricing model. Kara and Dogan (2018) study an inventory control problem of perishable items. The retailer is modelled as an RL agent, who uses the discretised remaining life of products on hand and inventory level as state information to decide the order quantity at the next period. The reward function takes into account both outdated and shortage costs. It is reported that SARSA performs better than QL and age information is helpful to agents.

There are research articles which study inventory control problems in manufacturing processes. Dhawan et al. (2009) study a problem which determines regular and emergency outsourcing decisions in an organisation. They propose a neuro-DP approach to deriving policies and improving actions, with the goal of minimising the expected total discounted cost. Their computational experiments show that the proposed neuro-DP approach performs slightly better than two other approaches, regular-mode-only and optimal-replenishment-model (Chiang, 2003). Wang et al. (2012) model an economic lot scheduling problem in factory, where a single machine can manufacture several products with different production times. They propose a tabular QL approach that captures the inventory levels of items and decides which items to produce, aiming to minimise long term costs. The authors develop an online learning approach, where a heuristic and RL are integrated to improve the solution quality at the beginning of the training. Their method sums the Q -values from both the heuristic and RL at the initial training steps, and gradually reduces the use of the heuristic. Ghorbel et al. (2015) use BayesiaLab, a software designed for Bayesian machine learning, to study an inventory problem of a manufacturing firm. The model aims to minimise the probability and damage of rupture (stock-out) of spare parts.

Recently, there is research on inventory control with a more practical and complicated setting. De Moor et al. (2021) study the inventory control problem of perishable goods with a standard DQN algorithm. Their reward is shaped such that there is an additional reward for the degree of similarity with well-performing heuristics policies. Meisheri et al. (2021) provide a DQN model for inventory control, taking into account the capacity constraints of trucks and multiple products. The reward includes fairness among different products, out-of-stock penalty, and wastage of perishing products. The DQN model learns the inventory level and predicted sales to produce a replenishment quantity as an action. In order to speed up computations while fulfilling the capacity constraint, the authors adopt a parallelised decision-making approach. Achamrah et al. (2021) propose an RL method for both SC and logistics purposes. For inventory replenishment, a threshold is used to facilitate a fixed size re-ordering policy. The shipping cost is minimised with an integrated approach of RL and GA, with operators such as crossover and mutations being the actions.

Table 14 summarises the applications for inventory control. Most of these problems have identical states, actions, and rewards. Therefore, the table focuses on the problem nature and learning methods.

Table 14
Summary of applications for inventory control..

Study	Method(s)	Reorder rule	Single/multi agent(s)
Ravulapati et al. (2004)	Tabular QL	Quantity-based	Multi agents
Dhawan et al. (2009)	Tabular QL	Quantity-based	Multi agents
Jiang and Sheng (2009)	Case-based RL	Quantity-based	Multi agents
Kim et al. (2010)	Regression	Safety lead time	Multi agents
Wang et al. (2012)	Tabular QL	Infinite supply	Single agent
Dogan and Güner (2015)	DQN	Quantity-based	Multi agents
Ghorbel et al. (2015)	Bayesian RL	Quantity-based	Single agent
Kara and Dogan (2018)	SARSA + DQN	Quantity-based	Single agent
Achamrah et al. (2021) ^a	DQN	Fixed policy	Single agent
De Moor et al. (2021)	DQN	Quantity-based	Single agent
Meisheri et al. (2021)	DQN	Quantity-based	Single agent

^aThe agent does not learn inventory replacement policy, but only evaluates routing.

Table 15
Summary of applications for pricing in supply chains.

Study	RL method	Action	Problem
Li et al. (2012)	DQN	Acceptance of orders & deliveries	Make-to-order production
Rana and Oliveira (2014)	$Q(\lambda)$ -learning	Pricing	Real-time pricing
Rana and Oliveira (2015)	$Q(\lambda)$ -learning	Pricing	Real-time pricing

3.4.2. Pricing in supply chains

Compared to the other two categories of studies in SCM, there are fewer studies whose focuses are on pricing. Li et al. (2012) study a make-to-order production system and consider the trade-off between price and lead time. They propose an RL approach with a two-layer perceptron, which acts as the agent to learn from order types, order quantities, and remaining production capacities. The perceptron then determines (i) whether to accept or reject an order and (ii) whether deliver or hold completed orders. If a production order is accepted, then price and lead time are chosen from the given sets. Rana and Oliveira (2014) consider a real-time dynamic pricing problem, where the firm has to sell inventories by a due-date given non-stationary demands. They propose a Q -learning algorithm approach which learns from the state-action pairs (remaining capacity and price) in an online environment. In a follow-up paper, Rana and Oliveira (2015) present two $Q(\lambda)$ -learning models with independent and interdependent learning of perishable goods/services prices. The Q -agent learns from the customer demands, selling prices, available stocks, and capacities to deduce the prices of different items. Table 15 summarises the applications for pricing in supply chains.

Our review of literature on pricing suggests that there requires further research on this topic, particularly related to the pricing decisions of the upstream parties in a SC. It is also worth studying how price variation influences the agents' actions in an RL environment.

3.4.3. Supply chain coordination

The studies reviewed in Sections 3.4.1 and 3.4.2 are mainly related to decisions of a single party or separate decisions of multiple parties in a SC. However, uncoordinated decisions which aim to optimise individual performance of each party may lead to suboptimality of the overall problem, or even adverse outcomes due to potentially conflicting objectives of different parties. To optimise the overall performance of the SC, coordinated decisions of multiple parties at different tiers would be essential. In this subsection, we review literature which applies RL for SC coordination problems.

Among all types of applications of RL for SC coordination, the most popular type is VMI. Similar to the studies in Section 3.4.1, literature on VMI focuses on inventory management problems. In VMI, vendors are responsible for the inventories of retailers, and hence may mitigate the risk of the Bullwhip effect (Lee et al., 1997a,b). The following articles examine VMI with RL methods. Their problem settings are similar, but methodologies vary.

Kim et al. (2005) provide a comparison between centralised VMI and a decentralised approach with two QL models. Suppliers or retailers learn the safety stock levels, below which an order is placed automatically. It is found that centralised coordination is better regarding both stability and cost. In a later paper, Kim et al. (2008) draw similar conclusions, where an asynchronous action-reward learning model is applied. Another paper by Kwon et al. (2008) propose a case-base myopic RL method for VMI with one supplier and four retailers. Similarly, Zarandi et al. (2013) study a VMI problem with one supplier and four retailers and propose a fuzzy RL algorithm to solve the problem. Gaussian membership functions are adopted in the model. The fuzzy RL model captures the on-hand inventory and average demand and determines the order quantities of a retailer in the next step. Vanvuchelen et al. (2020) apply PPO with Critic network to coordinate replenishment decisions of two shippers. The holding costs, back order costs, and shipment costs are simultaneously optimised given the historical inventory.

In contrast to end-to-end approaches to solving in VMI problems, some studies apply demand forecasting in a prior stage and use the predicted demands as inputs in their RL models. Kwak et al. (2009) adopt a situation reactive approach to solving a two-tier VMI problem. They apply exponential smoothing and moving average to forecast the future demand. Then, they decide the replenishment quantity, taking into account a compensation factor, predicted demand, and current inventory level. Sui et al. (2010) propose a

two-echelon VMI model with DQN adopted as the RL algorithm. The model uses the current inventory level and retailers' demand forecast to decide the number of trucks to dispatch for replenishment at each time period. In Afridi et al. (2020), a regression RL model takes the current forecast status (forecast error in the previous period), together with the forecasted demand, replenishment, and several forecast parameters, to determine replenishment orders. The reward is given as the deviation between the current stock level and the mean predicted stock level. Chien et al. (2020) propose a DRL approach that only considers demand forecasting but not inventory control in the semiconductor industry. A number of conventional forecast models and machine learning models – naïve forecast, simple moving average, single exponential smoothing, Syntetos–Boylan approximation, ANN, RNN, and support vector regression – are applied to forecast product demands. A DRL model is used to select the best demand forecast model for each product with its individual demand pattern.

In a typical VMI problem, parties involved are at two different tiers (e.g., suppliers and retailers). In SC coordination, a more general setting is a multi-tier structure where different players at different echelons *jointly* optimise the overall performance of the SC. In such a case, a greater number and more types of decisions, such as manufacturers' inventories, retailers' inventories and pricing, and perishable item inventory, are determined simultaneously.

To the best of our knowledge, Giannoccaro and Pontrandolfo (2002) and Pontrandolfo et al. (2002) are among the first to apply RL for inventory optimisation in a multi-tier SC coordination setting. Giannoccaro and Pontrandolfo (2002) consider a four-tier SC with a tabular QL approach. The agents learn from the current and historical inventory levels and then decide the order quantities. With only one agent per echelon, the common goal of agents is to minimise the transport, inventory, and waiting costs of the entire SC. Compared with a periodic order policy, this RL approach could lead to better SC performance. Pontrandolfo et al. (2002) consider the context of global SCM, where the parties involve production plants, export warehouses, import warehouses, and final markets in different countries. An agent decides whether to allocate a withdrawal order to an import/export warehouse and the transport mode (slow and fast carriers). Their proposed semi-Markov average reward technique (SMART) outperforms a local heuristic and a balanced heuristic. They conclude that an integrated coordination policy manages SC better and the proposed RL approach achieves a good trade-off between shifting production levels and smoothing of operations. In a later paper, Chaharsooghi et al. (2008) apply a similar method with the $Q(\lambda)$ ordering rule (Kimbrough et al., 2002), where $Q(\lambda)$ is the current downstream ordering size and Q is the action. In a similar setting, Mortazavi et al. (2015) propose a simulation-based optimisation framework with the use of QL. More detailed statistical measures, such as value at risk and coefficients of variation, are considered and a sensitivity analysis is conducted. In these papers, the agents represent decision makers in each tier, responsible for ordering decisions in its own echelon. Fuji et al. (2018) investigate a beer distribution game, where the SC parties include the factory, distributor, wholesaler, retailer, and customer. They propose a multi-agent RL method with DNN-weight evolution to solve the problem.

More recent research studies have solved similar problems, but with novel and more sophisticated RL methods and greater comprehensiveness of computational experiments. Peng et al. (2019) use a PG method to conduct a comparison of the approaches under continuous and discrete action space settings in a capacitated three-tier SC. They consider a centralised control approach, where both production target and delivery quantity are actions. In addition to a NN, they incorporate a post-processing unit to eliminate the infeasible actions and increase the chance of obtaining feasible actions. This algorithm is demonstrated to outperform a rule-based inventory control. Alves and Mateus (2020) present a four-echelon SC decision-making problem, where there are two parties per echelon. They propose a PPO, which is a PG approach, to determine the amounts of materials to be provided by a party and to be transported between parties. Gijsbrechts et al. (2020) study lost sales inventory model, dual sourcing model, and multi-echelon SC model with an A3C algorithm. In every scenario, a multi-layer perceptron agent takes the order quantity and inventory level of the previous time period and decides the order quantity at the next time period according to the softmax rule. For a multi-echelon SC, the dimension of actions is higher, and the upstream supplier is also capacitated. Their results suggest that the A3C method has better performance than constant order, base stock, and capped base stock policies. Oroojlooyjadid et al. (2021) propose a transfer-learning Shaped Reward Deep $(X + Y)$ Network (SRDQN) to study a four-tier beer game. Due to the complexity of multi-agent systems, only one of the players is set as the learning agent each time to train against base-stock and Stermann policy co-players. Each agent learns its own out-of-stock and storage costs and the cost of the entire system through a replay buffer. Their results suggest that SRDQN could significantly improve the learning party and reduce the systems' overall costs amongst Stermann co-players (a policy that mimics human behaviour), even though one of the non-learning players might worsen off. Alves et al. (2021) examine a four echelon SC system with full control of all suppliers. NN takes demands, stock levels, amounts of available materials and number of remaining steps as input to decide the shipment of materials from upstream to downstream.

Most of the existing studies assume either fixed order quantity and variable safety level, or variable order quantity. A recent work of Kosasih and Brintrup (2021b) shows that RL models can determine both order quantity and safety stock levels simultaneously in a typical SC. They provide a comparison of offline centralised and decentralised models with both QL and Advantage Actor Critic (A2C). Nurkasanah (2021) utilises a quantity-based Q-table approach for simultaneous optimisation of multiple parties in a three-echelon SC. The study considers the capacities of manufacturers and warehouses, minimum order limit, demand stochasticity, and lead time variation.

In addition to VMI and multi-tier SC coordination, there are further applications of RL in a SC context. For example, in manufacturing processes, production capacities and schedules are primary decisions which determine the timeliness of the delivery of products to respond to customer demands. Cao et al. (2003) study a two-agent RL-based SC optimisation model, with a production capacity constraint. The first agent uses the capacity and demand information of commodities to determine the optimum usage ratio of remaining components. The second agent learns the build plan (either postpone or advance the schedule of commodities), based on the customer demands. Aissani et al. (2012) present a dynamic scheduling problem in a SC which involves multiple production

Table 16
Summary of applications for supply chain coordination.

Study	Method(s)	Reorder rule	Problem	Single/multi agent(s)
Giannoccaro and Pontrandolfo (2002)	Tabular QL	Quantity-based	4-tier SCM	Multi agents
Pontrandolfo et al. (2002)	SMART	Quantity-based	Global SCM	Multi agents
Cao et al. (2003)	Tabular QL	Choice based	Production scheduling	Multi agents
Kim et al. (2005)	Tabular QL	Safety level	VMI	Multi and single agents
Chaharsooghi et al. (2008)	Tabular QL	Quantity-based	4-tier SCM	Multi agents
Kim et al. (2008)	Tabular QL	Safety level	VMI	Multi and single agents
Kwon et al. (2008)	Case-based RL	Safety level	VMI	Single agent
Kwak et al. (2009)	Regression	Quantity-based	VMI	Single agent
Sui et al. (2010)	DQN	Quantity-based	VMI	Single agent
Aissani et al. (2012)	SARSA	Quantity-based	Production scheduling	Single agent
Zarandi et al. (2013)	Fuzzy RL	Quantity-based	VMI	Single agent
Mortazavi et al. (2015)	Not mentioned	Quantity-based	4-tier SCM	Multi agents
Fuji et al. (2018)	AC	Quantity-based	4-tier SCM	Multi agents
Peng et al. (2019)	PG	Quantity-based	3-tier SCM	Single agent
Afridi et al. (2020)	DQN	Quantity-based	VMI	Single agent
Alves and Mateus (2020)	PPO	Quantity-based	4-tier SCM	Single agent
Chien et al. (2020)	DQN	Weight learning	Demand prediction model selection	Single agent
Gijsbrechts et al. (2020)	A3C	Quantity-based	2-tier SCM	Asynchronous agents with common objective
Puskás et al. (2020)	DQN	N/A	Vehicle dispatching	Single agent
Vanvuchelen et al. (2020)	PPO	Quantity-based	VMI	Single agent
Alves et al. (2021)	PPO, A2C, SAC	Quantity-based	4-tier SCM	Single agent
Chen et al. (2021a)	DQN	Quantity-based	3-tier SCM	Single agent
Kosasih and Brintup (2021b)	QL+A2C	Quantity-based + safety level	3-tier SCM	Multi and single agents
Nurkasanah (2021)	Tabular QL	Quantity-based	3-tier SCM	Multi agents
Oroojlooyjadid et al. (2021)	SRDQN	Quantity-based	4-tier SCM	Single agent

Table 17
Summary of typical characteristics of applications in supply chain management..

Category	State	Action	Reward
Inventory control	Inventories, capacities, and historical orders	Order quantities, safety stock levels	Individual holding and backorder costs
Pricing	Capacities and inventories	Acceptance and prices	Profits
Supply chain coordination	Inventories, capacities, and historical orders	Order quantities and safety stock levels	Total holding and backorder costs

sites. Due to the complexity of the problem, the authors propose an MARL framework, where there are an observer agent, a stock agent, and a resource agent at each site, to solve the problem. A SARSA algorithm is proposed to learn the optimal policy.

There are emerging types of SCs, powered by advanced technologies. RL has also been successfully deployed to tackle problems. Puskás et al. (2020) present a multiple platoon collaboration problem in a physical internet-based SC context. In this problem, the dispatcher, who is represented by an agent in their proposed RL framework, has to decide source–destination pair for a vehicle. A DQN algorithm is proposed to learn the actions. Their experiments show that the proposed RL approach perform better than heuristics. Chen et al. (2021a) present a blockchain-based agri-food SC to ensure reliable product traceability and safety. A DQN-based SCM model is proposed to determine the amount of production, storage, and shipment in order to achieve higher profits. The authors report that their proposed approach has higher and more stable rewards than *X*-table and heuristics under various scenarios.

Table 16 provides the summary of applications for SC coordination.

We summarise the typical characteristics in all SCM applications reviewed in this section by topic, as presented in Table 17. In Table 18, we present the problem settings of the applications in both inventory management and SC coordination.

From the literature review on SCM, we offer critiques of the applications in the following aspects.

Demand prediction and SC optimisation: Some studies directly optimise SC performance according to the inventory level and historical demands (e.g., Giannoccaro and Pontrandolfo, 2002), whilst some others adopt prediction models to forecast demands and then carry out optimisation (e.g., Kim et al., 2010). Our review finds that the mainstream of research in SCM does not focus on the prediction of demands, but performs optimisation under certain assumptions of demands (e.g., following a uniform distribution). Correlations of demands are rarely considered. On the other hand, optimal policies are derived according to specific demand patterns (e.g., Vanvuchelen et al., 2020).

Reordering rules: Most of the studies adopt quantity-based reordering rules (e.g., Alves et al., 2021; Vanvuchelen et al., 2020; Kwak et al., 2009); only a few articles consider safety stock level or lead time with a fixed reorder size (e.g., Kim et al., 2008, 2005). RL frameworks to implement these two types of reordering rules are quite different. A comparison of the two types of reordering rules would help understand their performances under different situations.

Production substitution: Only a few research studies consider multiple items and the possible substitution of a product by another (e.g., Achamrah et al., 2021). A typical setting of the existing literature is to assume that customer demands follow a

Table 18

Summary of problem settings of the applications in both inventory management and supply chain coordination.

Problem feature	Setting	Number of papers
Lead time	Constant	21/31
	Stochastic	10/31
Reorder rule	Quantity-based	25/31
	Safety stock	5/31
	Safety lead time	1/31
Reorder size	Fixed	5/31
	Variable	26/31
Capacity of supply	Infinite	22/32
	Limited	10/32
Demand distribution	Real customer data	7/39
	Normally distribution	9/39
	Uniform distribution	8/39
	Poisson distribution	6/39
	Other distributions	6/39

Notes:

- Other distributions include Erlang, exponential, and Gamma distributions.
- These figures in each subsection may not aggregate to 100%, because some articles consider multiple scenarios.

certain probability distribution. If the product is not available, the demand is not satisfied and lost. In some situations, customers may consider similar products to satisfy their needs. The modelling of such effects can be incorporated by learning from the historical data.

Integrated vs. sequential: RL approaches have a different methodological framework from a traditional two-phase approach of prediction and optimisation in SCM applications. One of the widely adopted types of RL algorithms is to integrate both prediction and optimisation procedures, considers all types of information (e.g., inventory levels and historical demands) as input, and directly determines the final solutions (e.g., reorder size) (e.g., [Vanvuchelen et al., 2020](#)). From our review, DNN is found to be the most popular method to model the RL agent. On the other hand, a sequential approach carries out prediction first (e.g., [Sui et al., 2010](#)). A separate RL agent considers the prediction as the state to determine decisions. This approach usually involves a regression prediction model. While an integrated approach is expected to be more effective in understanding better the end-to-end relationships between problem inputs and decisions, it may be worse off due to the challenge of achieving optimality in a more complicated setting. Further experiments would be essential to compare the integrated and sequential RL approaches.

Online vs. offline learning: An online RL algorithm can efficiently adapt to demand pattern variation but may suffer from low-quality initial solutions. It is a critical concern to the industry because poor actions adopted even only at the beginning would result in serious adverse consequences. In contrast, offline learning yields good results after sufficient iterations, but cannot adapt to changes. Researchers who apply regression approaches for demand prediction may adopt heuristics to generate good initial solutions to counter the negative effects of online RL algorithms ([Wang et al., 2012](#)). In addition, pretraining an online RL algorithm in a simulation environment is also a plausibly feasible direction.

Discrete vs. continuous action space: Most of the existing studies assume discrete order quantities or safety stock levels. It is proven by [Peng et al. \(2019\)](#) that continuous action space offers more effective solutions than the discrete counterpart under stochastic demands. However, the tradeoff between solution effectiveness and computational efficiency would require further research.

Centralised or decentralised decisions: In a SC context, decentralised decisions lead to suboptimality of overall SC performance due to separated decision-making. Unavailability of information shared among SC parties imposes further challenges, such as the bullwhip effect. In all the studies reviewed, not all SC parties are modelled as RL agents.

3.5. Reinforcement learning vs. conventional methods

RL is a relatively new framework for optimisation. Traditionally, optimisation problems in logistics and SCM are solved by exact methods, e.g., branch-and-cut algorithms, and heuristics (including meta-heuristics), e.g., Tabu search, simulated annealing, GA, iterated local search, and variable neighbourhood search. By solving problems with exact methods, global optimality of solutions is guaranteed; however, solution times are expected to be unacceptably long when solving complicated and large-scale real-world problems. On the other hand, heuristics offer fast solutions for practical use, but the solution quality may not have any theoretical guarantee. RL is widely considered a class of heuristics to approximate solutions, except that under certain problem assumptions it can ensure solution optimality. Compared with typical heuristics, which require pre-defined rules to guide solution search, RL learns from its reward function and previous actions to improve and explore solutions without the need of designing problem-specific rules. Thus, an advantage of RL is that complicated systems, which are hardly represented by solvable optimisation models or heuristic search rules, can be imitated by other ways of constructing the environments, for example, with simulators. Training time for an RL framework is often a main concern. In particular, many of the applications in logistics we reviewed are operational problems and require near real time solutions, where computing time is a dominating factor for the choice of solution methods.

Fortunately, once an RL model is trained, given the state, solutions can be delivered in seconds. Such an advantage lays out the popularity of RL, particularly in urban logistics (which we will have a discussion with a statistical analysis in Section 4.1). To demonstrate the performance of RL, most studies we reviewed compare their proposed approaches with exact methods and/or heuristics methods. Exact methods are mainly compared to assess solution quality, while heuristics are compared to demonstrate computational efficiency.

In TSP and VRP, RL is compared with cutting-edge exact methods and heuristics. For example, Kool et al. (2019) propose an RL model based on attention layers. On their computational instances of TSP and CVRP, their proposed attention model (with greedy decoding) is significantly more efficient than state-of-the-art exact solver Gurobi and heuristic solver LKH3 (reducing over 99% of the computing time while achieving an optimality gap of less than 8%) and can deliver the best solution among all compared heuristic solutions in a comparable solution time. For a detailed comparison of different RL algorithms and exact methods/heuristics, we refer the reader to Wang and Tang (2021). In their survey, various RL approximators, such as GCN, transformer, and RNN, are compared against exact solutions solvers and heuristics, including Gurobi, LKH3, nearest insertion, and nearest neighbour. They report that RL can constantly approach solutions to small-scale problems, and generally are much faster to deliver solutions of better quality, compared to standard heuristics.

In urban logistics, we find that, out of the 28 articles reviewed, three of them use MILP as benchmarks, (James et al., 2019; Chen et al., 2021b; Qiu et al., 2021), while 17 of them use heuristics as benchmark (Martin et al., 2016; Bono et al., 2018; Kang et al., 2019; Silva et al., 2019; Ahamed et al., 2020; Cao et al., 2020; Chen et al., 2020a; Ahamed et al., 2021; Ding et al., 2021; Hu et al., 2021; Li et al., 2021; Ma et al., 2021; Manchella et al., 2021; Mo and Ohmori, 2021; Qiu et al., 2021; Zou et al., 2021). Most of the benchmark heuristics are greedy or myopic heuristics, probably due to their simplicity to address the dynamic nature of certain urban logistics problems. This dynamic problem nature has also been observed in our review of RL applications for facility logistics. Those studies in dynamic facility logistics problems also tend to use greedy or myopic heuristics, such as first come first served and nearest first, while MILP is not used as benchmark. A representative example for conducting a performance analysis with MILP on dynamic problems is illustrated by James et al. (2019). The authors examine the performance of their RL approach on a dynamic PDP by constructing a series of static instances. In their experiments, MILP solved by Gurobi results in a computing time of around 6 h, whereas the proposed RL algorithm takes only 0.51 s. The authors also select several termination time points to examine the solution qualities offered by RL and Gurobi. They report that the solution delivered by RL (in only 0.51 s) has better quality than those by Gurobi at termination times of one minute and ten minutes. For static problems, Qiu et al. (2021) offer a comparison of their RL algorithm against LKH-D and adaptive large neighbourhood search algorithms. Their algorithm outperforms the benchmark heuristics in terms of computational time and solution quality on both small and large instances. In general, other studies also report performance improvements resulting from RL methods.

Similarly, in SC coordination, MILP, MINLP, and heuristics are adopted as benchmarks in most of the studies. Overperformance of RL over these benchmarks is generally reported in the reviewed studies. We remark that, in RL applications for inventory control, several research studies adopt existing replenishment policies for performance comparison. For example, Dhawan et al. (2009) consider two replenishment policies – optimal-replenishment-model and regular-mode-only – to assess the performance of their proposed neuro-DP model. Their approach contributes to an improvement over the benchmark replenishment policies, ranging between 0.9% and 1.54%. Some other studies in inventory control also adopt simple replenishment policies, such as fixed safety stock level and constant re-ordering size.

For a summary of the benchmark methodologies adopted in our reviewed studies for performance comparison, we refer the reader to Section 4.1. To conclude this subsection, research studies demonstrate in their applications that RL is significantly faster than exact methods to deliver solutions for practical use, and can achieve a higher solution quality than existing heuristics. Nevertheless, further research efforts in RL for logistics and SCM are required. We will provide an analysis of the current trends and propose some future directions in the next section.

4. Current trends and future directions

From our literature search, although a pool of methodologies and applications has been identified, there are still challenges to tackle the problems. In this section, we will present a statistical analysis of the reviewed studies to identify current trends and discuss future research directions.

4.1. Statistical analysis

Fig. 7 shows the numbers of studies on RL for logistics and SCM published from 2002 to 2021. We observe an increasing trend of publications, where the number of publications in the recent four years is around one and a half times the sum from 2002 to 2017. In particular, the number of studies on logistics is increasing more rapidly, probably due to the prevalence of the applications of RL for urban logistics research. The applications of RL are more inclined to SCM before 2016, but the applications in logistics have become dominant since the year due to the prevalence of E-commerce and growing demands for last mile delivery. We remark that, when we conduct this review, there are still upcoming publications in 2021 that cannot be included in this analysis.

Figs. 8 and 9 illustrate the popularity of different RL training methods in logistics and SCM. QL, including tabular QL and DQN, is the most popular approach to solving logistics and SCM problems, followed by AC (including A3C and DDPG) and PG. A small number of studies apply model-based or pure regression approaches. We remark that some studies use existing methods from other

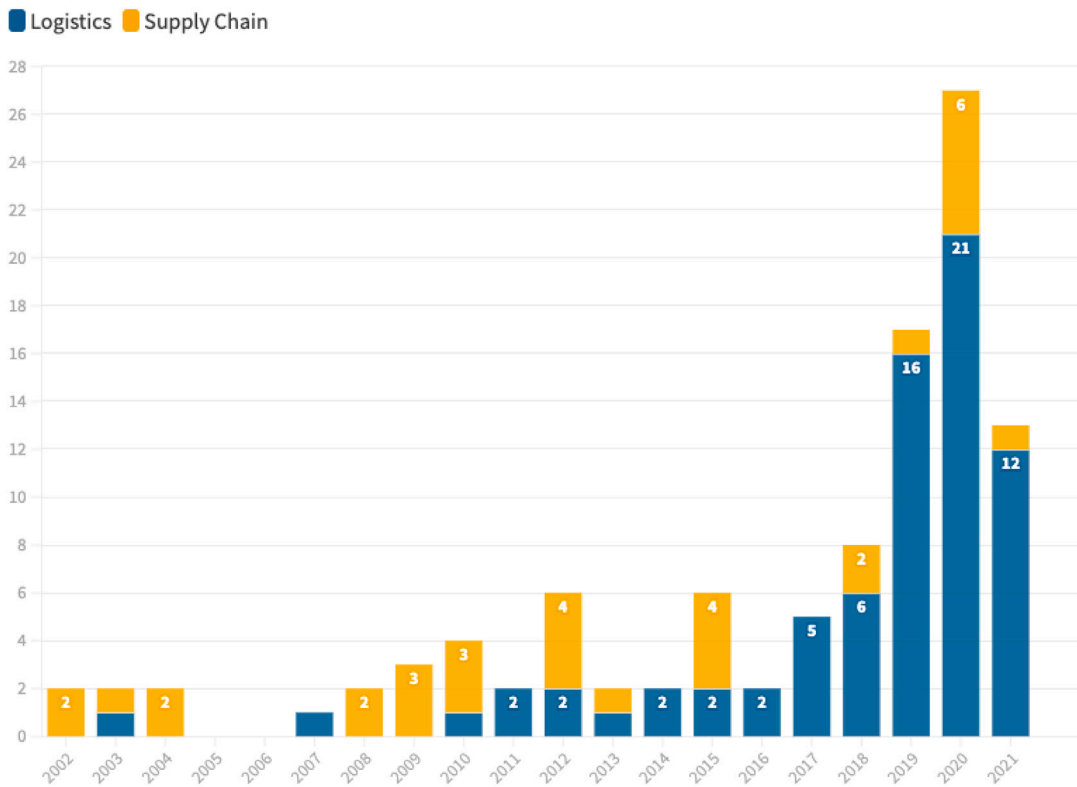


Fig. 7. Number of publications by year.

studies as benchmarks to conduct performance analysis; those used as benchmarks (not proposed by the authors) are not counted in this statistical analysis.

Fig. 10 shows the mapping of types of approximators adopted to applications. Multilayer perceptrons are the most widely used agents. Regression is the second most popular approach, particularly in SC problems due to its interpretability. Other advanced architectures such as GNN, RNN, and CNN are still rarely adopted, although they are prominent in solving combinatorial optimisation problems, such as TSP. We remark that, in Fig. 10, the sums of the numbers on the both sides may not equal because some studies develop multiple architectures. A detailed table of RL model(s) and approximator(s) by application domain is provided in the Appendix.

Fig. 11 provides the benchmark methodologies for performance comparison. Almost half of the studies apply heuristics, including greedy algorithm, GA, and DP, as the benchmark methodologies. Studies that construct baseline scenarios for performance comparison emphasise managerial insights, such as the effectiveness of VMI and decentralised inventory management, instead of algorithmic efficiency. Other benchmark methodologies for performance comparison include manual solutions and supervised learning solutions. Similarly, in Fig. 11, the sums of the numbers on both sides may not equal because some studies use multiple benchmark methodologies.

4.2. Future directions for methods

While our review of the state-of-the-art studies has demonstrated the successful applications of RL for logistics and SCM, technical challenges are also encountered and reported by some of the studies. We will present the challenges, which require significant efforts in future research.

Discretisation and continuity of state and action spaces: In our literature review, many of the RL applications for scheduling, pricing, and inventory management adopt continuous state and action spaces (e.g., Wang et al., 2018; van Heeswijk et al., 2019; van Heeswijk, 2020, 2021). However, the majority of learning methods developed requires discretisation of the action space, with a softmax rule applied to evaluate the probability of choosing an action. There are some studies that propose deterministic actions or parametric models, which do not require discretisation of the action space. However, these methods are not fully explored in SC and logistics settings. For example, it would be interesting to examine how the administrator's pricing in a continuous space would affect the pricing in a transport spot market.

Centralised and decentralised control methods: Both centralised and decentralised control approaches for optimisation problems have been applied in the reviewed studies, but decentralised control methods are not fully explored in different types

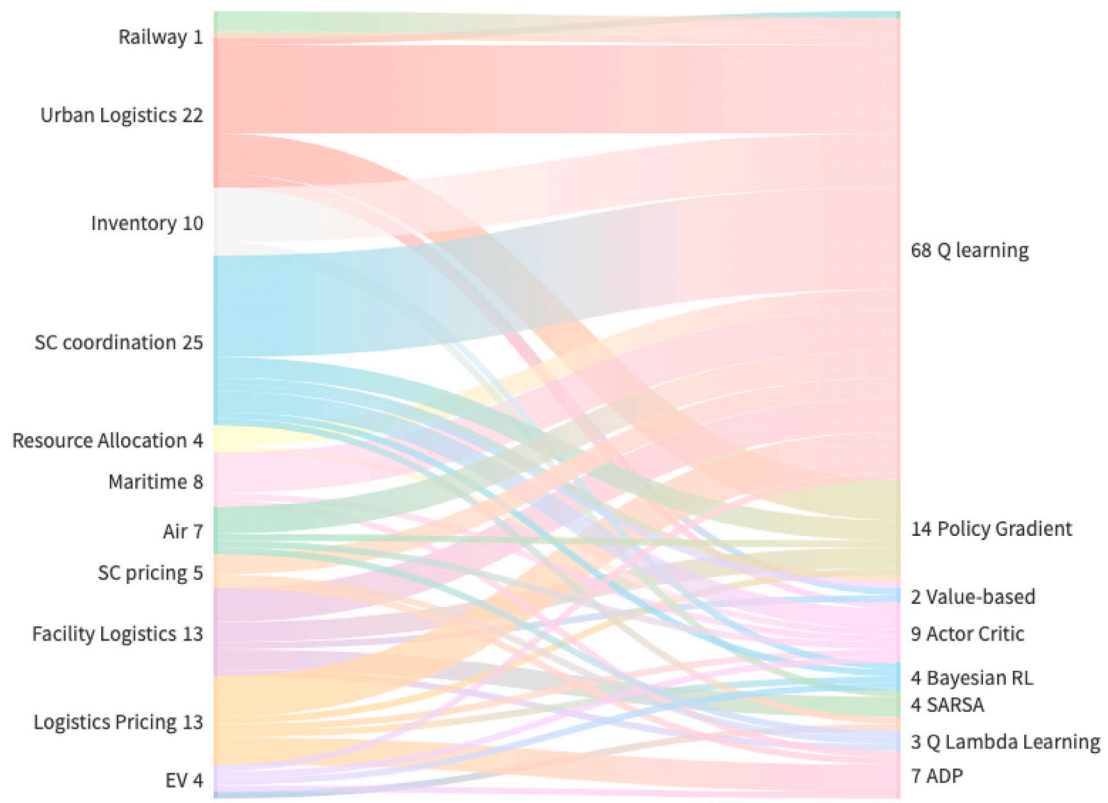


Fig. 8. Mapping of training methods to applications.

of problems. In a logistics context, multiple homogeneous agents, who represent tasks or items, can schedule or price themselves with a decentralised approach (e.g., van Heeswijk, 2020; Malus et al., 2020). In contrast, a centralised approach applies a single agent to make decisions (e.g., Qiu et al., 2021). This approach has been studied in a smart container context (van Heeswijk, 2020), but can be further extended to PDPs, as well as multi-hop transport. A common perception about decentralisation is that it may reduce system efficiency. For example, in a warehouse, without a centralised warehousing system, pickers may optimise individual routes, which may lead to interference at certain aisles. However, in the context of smart containers, the objective of the optimisation is to deliver the parcel at a minimal cost, where the price reflects the urgency of the delivery task. With this objective, the parcels may not be delivered in a selfish and myopic manner. This decentralised control approach can certainly reduce computational efforts in solving the problem.

Hierarchical reinforcement learning: Several reviewed studies on PDPs in Section 3.3.4 adopt RL for either assignment or routing, and then deploy a heuristic to solve the other problem. HRL, which adopts two levels of RL agents, offers an alternative to solving both problems in a single framework (e.g., Ma et al., 2021). It is an appropriate approach to delivery problems, where one agent is required to make assignment/acceptance decisions and the other can determine routing decisions. Similarly, in a SC context, the upper level of decisions can be the order size, while the lower level of decisions can be the scheduling of manufacturing jobs and the coordination of delivery tasks.

Adaptability of reinforcement learning models to system changes: For many offline RL algorithms, one set of model parameters may only be optimal to a particular problem setting. Once there is a variation in the system, one may need to retrain the model and calibrate the parameters from scratch. As discussed in Section 3.3.1, the scalability of RL methods is an issue. For example, in a PDP, if the number of vehicles changes, the parameters previously trained in an NN may not be applicable for the new problem anymore. Another example is the changes in demand patterns in SCM, which are commonly observed in the fashion industry (Choi, 2020b). An offline RL algorithm that trains the model from previous demand patterns may not be able to provide effective solutions to addressing the demands in the new season. New methodological frameworks are required to enhance the adaptability of algorithms to changes in the system, such that excessively repeated training can be avoided. One particular RL learning method that has not been adequately explored in our reviewed studies is fine tuning. While the calibration techniques for solving combinatorial optimisation problems have been studied, they are not yet applied to logistics and SC applications. Calibration techniques can reduce this burden by retaining most of the parameters of the NN and enhancing only a portion of the model parameters. However, the research on a comprehensive examination of this technique is still insufficient. For example, regarding varying demand patterns, one may compare this calibration approach against an online parametric RL model.

■ Q learning ■ SMART ■ Fuzzy Inference ■ Actor Critic ■ Bayesian RL ■ Policy Gradient ■ Regression ■ ADP
 ■ Value-based ■ SARSA ■ Q lambda learning ■ Inverse RL

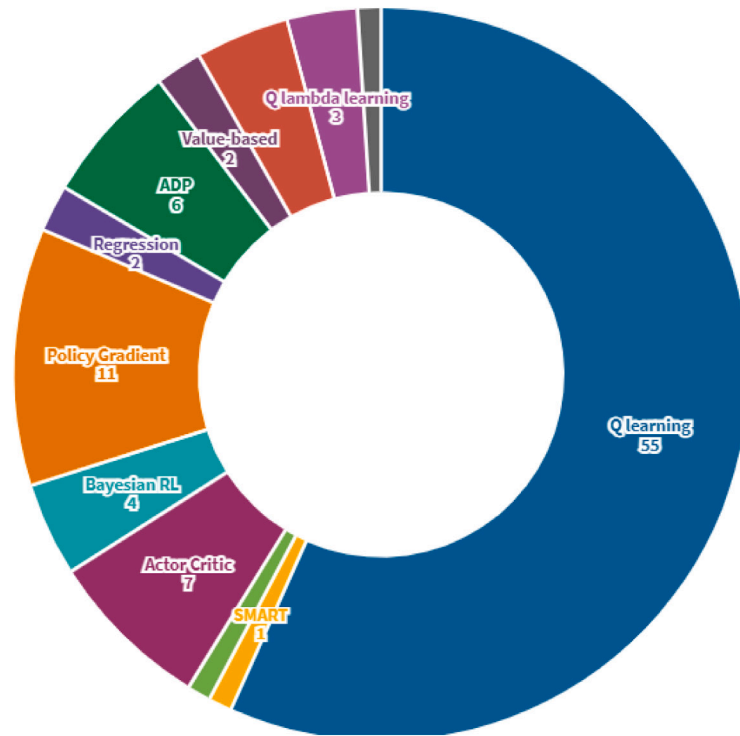


Fig. 9. Classification of training methods.

Comprehensiveness and fairness of performance comparison: In many studies, the proposed RL methods are compared against simple algorithms, such as greedy heuristics, or the existing practices in their cases studies. Such a comparison may not be comprehensive enough to claim the best performance among all existing methods. Furthermore, in these studies, the computational experiments are mostly constructed based on a particular case study or problem setting. The outstanding performance of a certain RL algorithm may be subject to specific objectives, scenarios, problem sizes, and computational environments. Nevertheless, conducting a comprehensive and fair performance comparison is challenging, as the implementation of benchmark RL algorithms may not be transferable to another environment. There is an urgent need to develop a unified and effective methodological framework for comparison purposes (Farazi et al., 2021).

Explainable AI: Explainable AI is gaining popularity in various industries. For example, in the European Union, rejection of a bank loan must be accompanied by a valid reason. A traditional black-box NN is unable to provide reasons for acceptance/rejection decisions, and thus explainable AI methods to assist loan application decisions have become more popular (Sachan et al., 2020). Similarly, in a SC or logistics setting, black-box models suggesting reorder quantities, delivery/production schedules, resource allocation, and task acceptance/rejection decisions may not be interpretable to human beings (Boute et al., 2021). Valid reasons for such decisions to be provided to other parties involved are, therefore, unavailable; trust between parties for long-term relationship may be more difficult to build. On the other hand, transparent regression models rely on rigid modelling assumptions. They may not be flexible as NNs, especially under the situation that non-linear relationships between factors and variables exist. A promising research direction would be to develop RL frameworks that deliver high-quality solutions and easily interpretable results. While we have not identified any research article that attempts to develop explainable AI in an RL framework, we do notice that the concept is already applied in domains other than RL. For example, Kosasih and Brintrup (2021a) propose a supervised learning approach with GNN to predicting the hidden links between suppliers.

Inverse reinforcement learning: IRL, which derives the reward function of an RL framework via observations of an agent's behaviour over time (Ng and Russell, 2000), is another promising research field. Recently, IRL has been applied to routing problems (Liu et al., 2020). Nevertheless, there is no doubt that IRL can be applied to other problems in the logistics and SC domains. For example, as discussed in Section 3.3.4, it would be interesting to apply IRL to learn from an agent's behaviour of the acceptance/rejection of pickup and delivery orders with various profits, in different times and regions, from historical data and infer the agent's reward function. With IRL, the priorities of pickup and delivery orders may be determined. This distinct feature of IRL from other RL algorithms has contributed to its effectiveness in offering practical solutions that can address the real needs

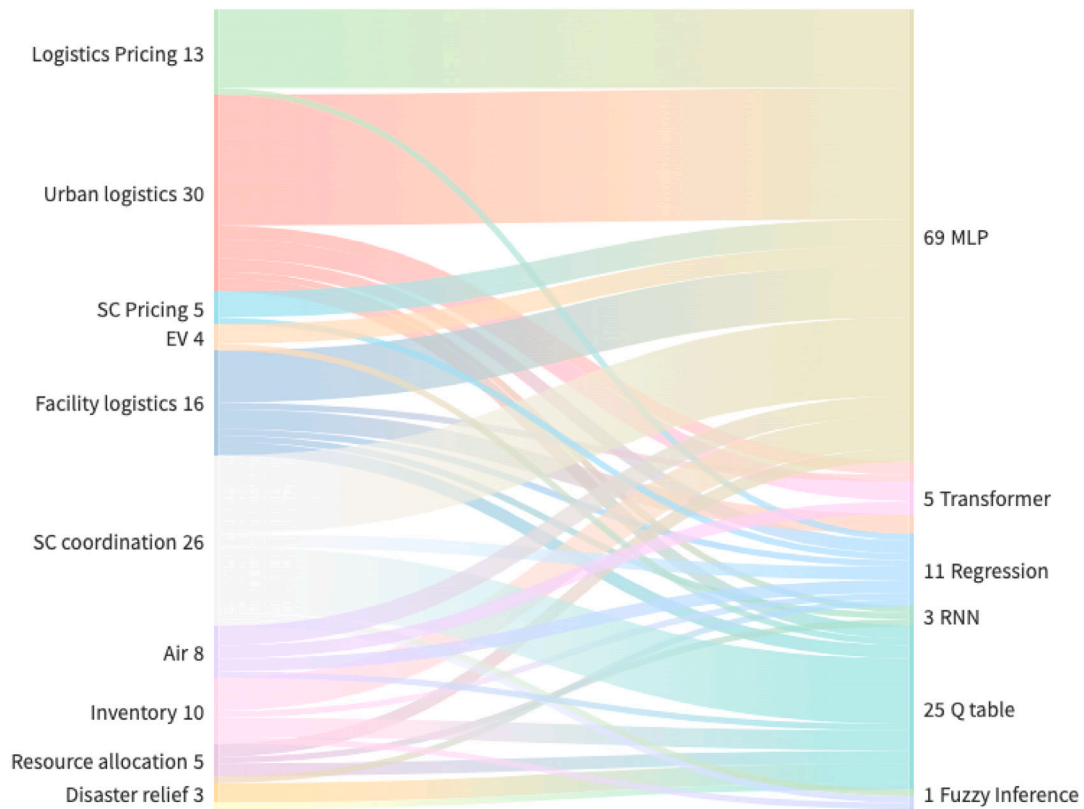


Fig. 10. Mapping of approximators to applications.

of decision-makers. Such a benefit would lead to a promising research direction that augments the popularity of applying RL for real-life problems and its actual implementations.

Partially observable states: One of the computational challenges of RL is its ability to handle partially observable states because agents would need certain memories to learn the behaviour (Singh et al., 1994; Toro Icarte et al., 2019). We have found only a small amount of research in relevant application fields, such as disaster relief, pricing, and SCM (e.g., Chen et al., 2021b). It would be interesting to explore the effects of different settings of unobservable states and offer performance comparison against heuristics.

Distributional reinforcement learning: In our reviewed studies, most of the RL frameworks consider the predicted reward a single value. Instead of a single value, Bellemare et al. (2017) model the output of a value function as a distribution. Their proposed distributional bellman operator discretises the value space with predefined intervals, and then evaluates the probability of the reward being within an interval during the learning process. This method has shown significant advantages in sparse reward problems. It may also have the potential to enhance the robustness of the system, as the predicted reward range can be obtained.

Reinforcement learning for robust decisions: In RL, the objective is to maximise the total expected reward, i.e., MDP (1). This objective is appropriate for determining optimal decisions in the long run (i.e., the decision process lasts for a sufficiently long period of time). However, in some applications such as disaster relief, decisions are used for only temporary purposes and the adverse outcomes can be severe under rare scenarios. In logistics and SC applications, robust optimisation has been proposed (Bertsimas and Thiele, 2006; Ardestani-Jaafari and Delage, 2018; Wang et al., 2021). Robust optimisation is powerful to offer good-quality solutions to hedge against the risk of encountering worst-case scenarios and uncertainty of probability distributions governing the system. However, in an RL framework, the challenge is to develop an effective way to incorporate solution robustness into the learning process, such as robust MDPs (Nilim and El Ghaoui, 2005; Wiesemann et al., 2013; Ho et al., 2018) and risk-averse MDPs (Delage and Mannor, 2009; Behzadian et al., 2021).

Utilisation of real data: DRL can extend its state space by incorporating real data to model the reality (Boute et al., 2021). In a SC setting, the state space may include traffic and weather conditions, so as to offer more predictions of the demand and lead time (Boute et al., 2021). In the logistics context, real data can be used for a more accurate prediction of the arrival time. The information can either be derived in a supervised learning approach, or through an end-to-end framework directly. One of the major challenges of the utilisation of data in practice is the integration of data from multiple parties in a SC, as there may be no incentive for them to share their data with the others and the data format and collection frequency may be different (Kuo and Kusiak, 2019). An other important aspect for effective logistics and SCM is digital interoperability (Pan et al., 2021a,b), which aims to enable reliable data

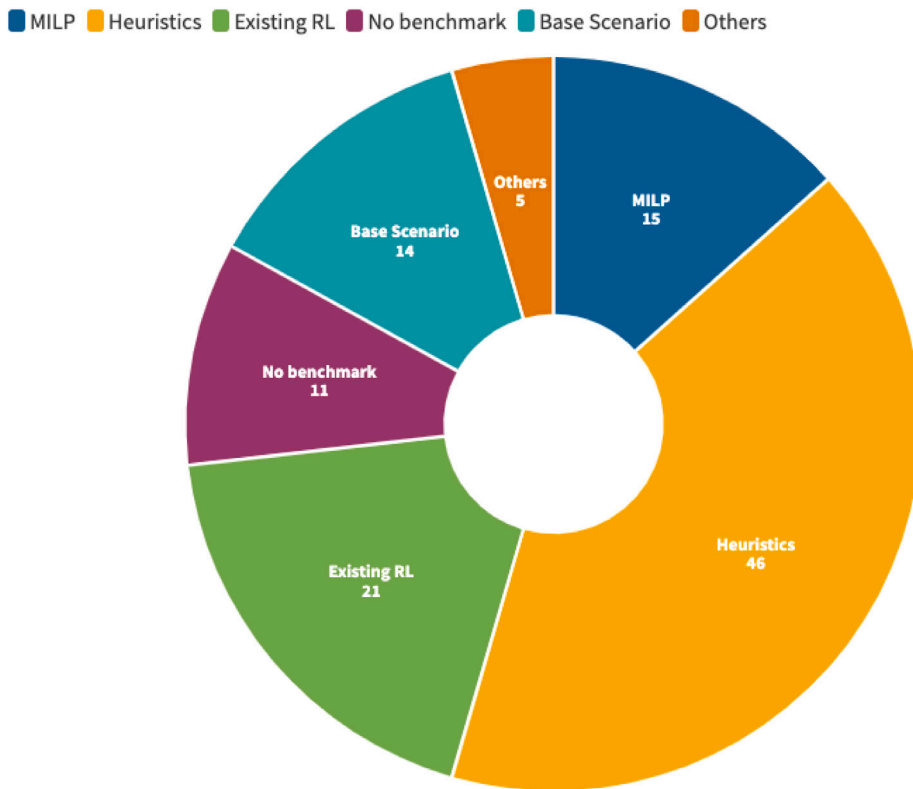


Fig. 11. Benchmark methodologies for performance comparison.

sharing among different types of sensing and computing devices across different systems in various organisations with the goal of strengthening cooperation or cooptation of the independent SC parties.

Generalisation: Most of the research studies in logistics consider a fixed number of delivery destinations. [Lisicki et al. \(2020\)](#) show that the generalisability can be improved by curriculum learning. In their study, approximators are trained with maps of increasing complexity over the training process. This would be an interesting direction to explore.

Online and offline training methods: An offline RL algorithm is trained with an environment prior to inference. During the inference phase, the parameters keep constant. However, the environment may not always remain stationary. The outcomes of an action and their state transition probabilities may vary from time to time. An offline algorithm does not have exploration and therefore cannot observe states, actions, and rewards that are not explored by the incumbent policy. On the other hand, an online algorithm explores all the time, but suffers from an exploratory cost. In recent years, researchers have proposed the concept of *conservative* RL. It adopts pre-trained RL parameters, but also updates their values. The additional objective of such learning method is to maintain the exploration cost to be within a certain budget with a high probability ([Xu et al., 2021](#); [Kumar et al., 2020](#)). To the best of our knowledge, this has not been applied to any logistics or SC contexts. It may be, for example, used for online inventory control policies where the demand patterns change over time.

4.3. Future directions for applications

In this subsection, we present some future directions of RL applications in logistics and SCM.

Environmental and sustainability concerns: Environmental and sustainability concerns include problems related to the reduction of carbon emission in transportation and wastes due to unsold perishable items. We find that only around 12% of the articles consider environmental and sustainability factors in their objectives or constraints ([Irannezhad et al., 2017](#); [Wangapisit et al., 2014](#); [Firdausiyah et al., 2019](#)). With the introduction of carbon pricing and waste charging in many countries, the incorporation of environmental and sustainability factors in the objective is increasingly important.

Perishability of goods The majority of the literature assumes goods are not perishable. Even perishability is considered, it is typically modelled in an aggregate scale (i.e., average storage duration) ([Kara and Dogan, 2018](#)). The challenges to incorporate the shelf lives or freshness of individual items in an RL framework arise from the higher dimensionalities of states and actions. An efficient RL framework is necessary for practical solutions.

Transferability of solution methods and results: Most of the reviewed studies rely on strong assumptions to develop their solution methods. For example, the RL framework and results may be applicable to only certain customer utility functions, demand

distributions, and policies. If these assumptions and settings do not hold, the solution method and results may no longer be valid. Therefore, transferability of solution methods and results is important, particularly when other researchers may wish to borrow the solution method, conduct numerical experiments of some problem variants, or quote the insights and conclusions from the study. The existing studies can serve as a good foundation for further research, such that researchers may attempt to relax modelling assumptions for the development of more general solution frameworks, derivation of promising conclusions, and delivery of more realistic solutions to address the practical needs (Farazi et al., 2021). The other aspect of transfer learning is to utilise existing studies to guide RL learning. For example, De Moor et al. (2021) demonstrate that using a well-defined heuristic to guide training can improve the performance and achieve better efficiency. Relevant studies remain very few at this moment.

Heterogeneity of resources: In many of the reviewed studies in logistics, fleets of vehicles and couriers for delivery tasks are homogeneous (e.g., Kool et al., 2019; Kalakanti et al., 2019; Joe and Lau, 2020). In practice, logistics companies often operate multiple types of vehicles (e.g., characterised by their sizes, speeds, and equipment) and couriers (e.g., characterised by their transport modes, delivery preferences, and working hours) to complete different delivery tasks (Zong et al., 2021). Similarly, in other logistics applications, such as those in facility logistics, container transport, and railway systems, resources are also heterogeneous in practice. The modelling of heterogeneity of resources can offer more practical solutions but requires establishing multiple types of RL agents; the effective modelling of the interactions between the RL agents, and the required efficient computations would need further research efforts.

Spatial-temporal information In RL models for logistics applications, temporal information is usually not fully utilised. For example, in a dynamic delivery problem, whenever there is a new user request, the delivery route can be re-planned accordingly such that advanced planning regarding the delivery schedule is less important. Meanwhile, RL approaches for SCM typically does not take spatial information into account, as the RL agent focuses on the historical order quantities. In recent research of ride-sharing applications, logistics service providers may also deploy idling personnel or vehicles to certain regions to anticipate potential future demands (Zong et al., 2021). To fulfil such an objective, the RL agent should know not only the latest information about the pending orders, but also historical data about the order demands. The fusion of historical demand data and spatial information has been proposed by a number of studies for applications such as traffic prediction, ride-sharing, and pricing (Li et al., 2018; Lin et al., 2019; Lu et al., 2019b). On the other hand, only a small amount of research on the utilisation of spatial-temporal information has been done in the logistics and supply chain domains. In many of our reviewed applications, the outputs of the RL frameworks are actions instead of demand prediction, the agents have to learn the spatial-temporal embeddings indirectly from the outcomes of its actions; such a learning approach may suffer from numerical instability.

Additive manufacturing: In our reviewed studies related to production, RL applications mainly focus on production scheduling, demand forecasting, and inventory management in traditional production settings (e.g., Wang et al., 2012; Ghavamzadeh et al., 2015; Peng et al., 2019; Chien et al., 2020). An emerging mode of production is additive manufacturing. As the technology has become more economically viable (Guo et al., 2021b), the ways of how companies and organisations operate their SCs are expected to be unconventional. Additive manufacturing facilitates mass customisation, decentralised production, inventory reduction, and savings from custom tariffs. It is particularly well suited to areas such as fashion and aircraft maintenance. In the fashion industry, customer preference and personalisation are key factors to success and both require product customisation. Aircraft maintenance requires massive volumes of spare parts to be stored in the warehouse for maintenance purposes. Nevertheless, none of the reviewed studies reviewed considers 3D printing in their research. It would be exciting to explore how RL can facilitate additive manufacturing, such as the optimisation of the inventory level of raw materials for 3D printing and the locations of mobile printing hubs.

Digital twin-enabled logistics and supply chain applications: Most of the RL frameworks reviewed in this paper rely on a simulation model to represent the environment (e.g., Mo and Ohmori, 2021; Zeng et al., 2011). Recently, digital twin technologies, which construct real-time virtual representations of physical assets or systems to model the physical interactions in the digital space (Mikell and Clark, 2018), have become mature and gained significant adoptions in various types of real-world applications (Kuo et al., 2021). Enabled by the Internet of Things, the real-time physical system state can be captured. Applications which leverage digital twin technologies to incorporate the real environment for the RL agents to interact with are expected to provide more effective and realistic model training and practical decisions.

Applications under pandemics: The outbreak of the Covid-19 pandemic has substantially transformed the modes of businesses and organisations, and the living behaviours of human beings. In particular, the operations of global and local SCs and international and urban logistics have evolved during this pandemic (Choi, 2020c; Mitrega and Choi, 2021). For example, delivery services have been brought closer to customers (Choi, 2020a; Choi and Sethi, 2021), where delivery resources are expanded and their allocation is optimised to address the vastly increased demands for last mile delivery services. SC infrastructures have been designed to be more robust and responsive to surges in demands and disruptions in operations. From our literature search, the research on RL for logistics and SCM under a pandemic situation is still lacking. There is great research potential to develop RL frameworks to support logistics and SC decisions under a pandemic, due to the adaptability of RL to uncertain and dynamic environments. For example, in an urban delivery problem under a pandemic, when delivery resources are constrained, priorities may be given to certain types of customers (e.g., senior citizens and other people in need). RL approaches would be effective in learning the preferences in customer prioritisation and order acceptance/rejection and optimising the trade-off among multiple objectives, such as maximisation of social welfare and customer satisfaction and minimisation of delay in delivery and operational cost. Another important topic under a pandemic is the distribution of vaccines. Bae et al. (2021) discuss the challenges of vaccine roll-out, including production limit, prioritisation, distribution with cold chain, fairness, and record keeping. There are a few studies that aim to address the fairness in the distribution of vaccines at a strategic level (Lim et al., 2019; Enayati and Özaltn, 2020). At the tactical and operational levels,

research is still missing. There is good potential to apply RL to balance the objectives, such as reduction of logistics and warehousing costs and enhancement of the protection rate for the population.

Integrated logistics and supply chain decisions: In existing research studies, logistics and SCM problems are typically studied separately. For example, in an inventory management problem, parties in a SC aim to determine production and order quantities. Meanwhile, in a logistics problem, delivery orders and product availability are given, while the vehicle routing has to be determined. The only article that considers both uses a fixed policy with no learning for inventory management (Achamrah et al., 2021). Solving the two problems independently may not be able to incorporate inter-dependencies of their parameters and decisions, which have been observed in practice. For example, the optimal production quantity may be constrained by truck capacities. Production schedules may also be affected by the availability of trucks. A potential future research direction is to assess further benefits by an integrated optimisation of SC and logistics decisions. However, solving such an integrated problem is challenging with an RL approach in the sense that (i) the two types of problems may adopt different state-action spaces (potentially having one continuous and the other discrete), (ii) the different types of decisions may require heterogeneous agents trained by multiple learning models, who would have further interactions between each other, and (iii) substantially higher complexities due to the coupling of the state-action spaces.

5. Conclusion

In this paper, we provide an introduction to RL methodologies and review their applications in logistics and supply chain management. We classify and review these studies by application, and discuss their challenges. The current trends and future directions in both methodological and application aspects are then presented. To conclude this paper, we highlight the strength and limitations of RL.

To conclude this section, we summarise the strength and limitations of RL.

Strength of RL: Compared with standard heuristics, RL has its unique strength. RL algorithms memorise a set of parameters such that actions can be determined promptly once the current state information is available. On the other hand, heuristics do not learn from previous actions but rely on pre-defined rules. In addition, RL may learn from historical data for prediction purposes and offer an integrated forecasting and optimisation approach to delivering solutions to stochastic problems in logistics and SCM, such as inventory control and disaster relief. On the contrary, heuristics rely on standalone forecasting models for prediction, and the optimality gap may accumulate during the solution procedure. Besides, a natural advantage for RL algorithms is the requirement for an environment to interact with. The majority of applications in logistics and SCM rely on simulation environments, where modelling assumptions are required.

Limitations of RL: The current RL algorithms face challenges when handling complex multi-agent systems. For example, under the topics of pricing, bidding, and negotiation in logistics, and supply chain coordination, multiple parties are involved in the decision-making process, such that different types of RL agents are required. The training process involving a large number of agents often yields solutions away from Nash equilibria, if they exist. One typical approach that we find from our reviewed studies is to reduce the number of agents in a system, thereby shrinking the action space. For example, in a facility logistics setting, instead of considering the vehicles, Hwang and Jang (2020) use junction nodes as the agents. However, simplifying the learning agents (e.g., imposing heuristic rules in the environment to guide decisions not determined by the RL agents) may worsen the solution quality and require building a more complicated simulation environment. For example, some studies apply greedy heuristics to assign tasks to agents. Moreover, RL algorithms are highly computationally expensive, where time spent on training is extensive. Since most RL algorithms are problem-specific, it is challenging to transfer the architecture and model parameters from one problem to another. To extend the applications of RL for logistics and supply chain problems, the algorithms should have strong adaptability and generalisability. Last but not least, as discussed, RL may not work well when the states in the environment are partially observable.

In conclusion, RL is a powerful tool to deliver solutions promptly once the model is well trained, particularly in cases where problem characteristics remain and historical patterns continue to exhibit. Nevertheless, there are still considerable research opportunities in logistics and supply chain management, regarding the required methodologies and further applications.

CRedit authorship contribution statement

Yimo Yan: Formulation of research questions, Overall design of review framework, Introduction to the methodologies, Review of literature, Result analysis, Manuscript writing. **Andy H.F. Chow:** Formulation of research questions, Overall design of review framework, Introduction to the methodologies, Review of literature, Manuscript writing, Supervision. **Chin Pang Ho:** Formulation of research questions, Introduction to the methodologies, Review of literature, Manuscript writing, Supervision. **Yong-Hong Kuo:** Formulation of research questions, Overall design of review framework, Introduction to the methodologies, Review of literature, Result analysis, Manuscript writing, Supervision. **Qihao Wu:** Introduction to the methodologies, Review of literature, Result analysis, Manuscript writing. **Chengshuo Ying:** Introduction to the methodologies, Review of literature, Manuscript writing.

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Appendix A. Reinforcement learning models and approximators by application domain

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